
Climate Monitoring & Insights Dashboard

*A Global Weather Intelligence System Built on the
Global Weather Repository Dataset*

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Tool	Python · Plotly · Streamlit
Dataset	Global Weather Repository (Kaggle)

This project covering data preparation, statistical analysis, and interactive dashboard development for global weather intelligence across 200+ countries.

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1. Introduction

1.1 Project Background

Weather and climate data represent one of the richest sources of real-world, multi-dimensional information available for data analysis. Understanding global temperature trends, precipitation patterns, wind behaviour, air quality, and extreme event frequencies is critical for public health planning, agriculture, infrastructure management, and environmental policy.

This project, developed under the **Infosys Springboard** program, builds a *Climate Monitoring & Insights Dashboard* — a fully interactive, web-based application that enables users to explore global weather patterns across more than 200 countries using data from the **Global Weather Repository** available on Kaggle.

1.2 Objectives

The project aims to achieve the following goals:

- 1. Data Preparation:** Acquire, inspect, clean, and transform raw global weather data into an analysis-ready format.
- 2. Statistical Analysis:** Identify distributions, correlations, seasonal trends, and extreme weather events across countries and years.
- 3. Visualisation Design:** Select appropriate chart types (choropleth maps, violin plots, radar charts, heatmaps, etc.) and design an intuitive dashboard layout.
- 4. Dashboard Development:** Build an interactive Streamlit + Plotly dashboard with filters, sliders, and region selectors, providing actionable climate intelligence.

1.3 Dataset Description

Table 1.1: Global Weather Repository — Dataset Overview

Attribute	Details
Source	Kaggle — Global Weather Repository
Granularity	Daily weather observations per location
Geographic Span	200+ countries worldwide
Time Coverage	Multiple years (multi-annual records)
Key Variables	Temperature, Precipitation, Wind Speed, Humidity, Cloud Cover, UV Index, PM2.5, PM10
Output Format	CSV (daily and monthly aggregated)

1.4 Project Milestones Overview

The project was structured into three milestones over six weeks:

Table 1.2: Project Milestones Summary

No.	Milestone	Key Deliverables
M1	Data Preparation (Weeks 1–2)	Cleaned daily and monthly CSVs; unit conversions; missing value handling
M2	Core Analysis & Visualisation Design (Weeks 2–4)	Statistical summaries; correlation analysis; extreme event identification; chart selection
M3	Dashboard Development (Weeks 4–6)	8-tab interactive Streamlit dashboard; Plotly charts; KPI cards; global maps

2. Proposed System

2.1 Overview

The proposed system is a **Climate Monitoring & Insights Dashboard** — a fully interactive, web-based data intelligence application built using Python, Plotly, and Streamlit. It is designed to transform raw, multi-dimensional global weather data into actionable climate insights through a structured pipeline of data preparation, statistical analysis, and interactive visualisation. The system replaces static reports and manual data queries with a dynamic, real-time-filterable dashboard accessible through any web browser.

2.2 System Architecture

The system follows a layered architecture consisting of four interconnected components:

- 1. Data Layer:** Raw CSV data from the Global Weather Repository is ingested, cleaned, and transformed into two structured outputs — a daily cleaned dataset and a monthly aggregated dataset. These serve as the sole data sources for all downstream analysis.
- 2. Analysis Layer:** Statistical computations including distributional analysis, correlation matrices, seasonal trend extraction, extreme event detection, and composite risk scoring are performed on the prepared data using `pandas` and `numpy`.
- 3. Visualisation Layer:** Plotly Express and Plotly Graph Objects render 40+ interactive charts — including choropleth maps, violin plots, radar charts, polar charts, gauges, and treemaps — each tailored to communicate a specific analytical insight.
- 4. Presentation Layer:** Streamlit provides the web application framework, organising the dashboard into 8 thematic tabs with a shared sidebar offering real-time country and year filters. Caching via `@st.cache_data` ensures responsive performance during interactive exploration.

2.3 Key Features of the Proposed System

Table 2.1: Proposed System — Key Features

Feature	Description
Interactive Filtering	Real-time country and year selection via sidebar dropdowns; all charts and KPI cards update instantly
8-Tab Dashboard Layout	Thematic tabs covering Overview, Temperature, Precipitation & Wind, Extreme Events, Regional Comparison, Humidity & Cloud, UV & Air Quality, and Climate Risk Index
Composite Risk Score	A formula-driven Climate Risk Score (0–100) integrating temperature, precipitation, wind, UV, and PM2.5 thresholds into a single hazard metric
Global Choropleth Maps	Four world maps encoding Temperature, Precipitation, Air Quality, and Risk Score geographically
WHO Benchmarks	PM2.5 and PM10 reference lines derived from WHO air quality guidelines displayed on relevant charts
Extreme Event Detection	Threshold-based flagging of Heatwave, Heavy Rain, High Wind, Cold Stress, High UV, and Poor Air Quality events
Year-over-Year Anomaly	Automatic computation and display of temperature deviation between the selected year and the prior year
Performance Caching	Memoised aggregation functions prevent redundant computation on repeated filter interactions

2.4 Comparison with Existing Approaches

Traditional approaches to weather data analysis typically rely on static spreadsheets, pre-rendered PDF reports, or specialised GIS tools requiring expert knowledge. The proposed system addresses the limitations of such approaches as follows:

Table 2.2: Proposed System vs. Existing Approaches

Dimension	Existing Approaches	Proposed System
Interactivity	Static charts, no real-time filtering	Dynamic Plotly charts with hover, zoom, and sidebar filters
Accessibility	Requires GIS or BI tool expertise	Browser-based; no installation required for end users
Data Coverage	Single-country or single-variable focus	200+ countries, 8+ weather variables in one platform
Risk Quantification	Qualitative severity descriptions	Composite numeric Climate Risk Score (0–100)
Air Quality Benchmarking	Manual threshold comparison	Automated WHO guideline reference lines on all AQI charts
Performance	Full reload on each query	Cached aggregations for near-instant filter response

2.5 System Workflow

The end-to-end workflow of the proposed system proceeds as follows:

Step 1. Data Ingestion: The raw `GlobalWeatherRepository.csv` file is loaded into a pandas `DataFrame`.

Step 2. Data Cleaning: Redundant imperial-unit columns are dropped; missing values are imputed using median (numeric) and mode (categorical) strategies; all measurements are converted to SI units.

- Step 3. Feature Engineering:** Temporal features (year, month, day) are extracted; SI-unit columns (wind in m/s, pressure in Pa, visibility in metres) are computed; daily records are aggregated to monthly averages per country and location.
- Step 4. Analysis:** Distributional, correlation, seasonal, regional, and extreme-event analyses are performed on the cleaned dataset.
- Step 5. Risk Scoring:** A vectorised Climate Risk Score is computed for every record using the composite formula, then averaged per country and month.
- Step 6. Dashboard Rendering:** The Streamlit application reads the cleaned monthly CSV, applies sidebar filter selections, executes cached aggregations, and renders all charts and KPI cards within the relevant tab.
- Step 7. User Interaction:** The user selects a country and year from the sidebar; all charts update reactively without a page reload.

2.6 Tools and Technologies

Table 2.3: Tools and Technologies — Proposed System

Tool / Library	Version	Role in System
Python	3.10+	Core programming language
pandas	2.x	Data loading, cleaning, and aggregation
numpy	1.x	Numerical operations and vectorised risk scoring
Plotly Express	5.x	High-level interactive chart generation
Plotly Graph Objects	5.x	Low-level chart customisation and subplots
Streamlit	1.x	Web application framework and tab layout
matplotlib	3.x	Static exploratory charting during analysis
seaborn	0.13+	Statistical visualisation (heatmaps, box plots)
Google Colab	—	Cloud-based development and execution environment

2.7 Expected Outcomes

The proposed system is expected to deliver the following outcomes upon completion:

- A fully functional 8-tab interactive dashboard deployable via `streamlit run` with no additional configuration.
- Two cleaned, analysis-ready CSV files (daily and monthly) suitable for further research or integration with other tools.
- A quantified Climate Risk Score for every country in the dataset, enabling prioritisation of high-risk geographies.
- WHO-benchmarked air quality analysis for over 200 countries, directly supporting

environmental health monitoring.

- A reusable, modular codebase structured around clearly separated data preparation, analysis, and visualisation components.

3. Methodology

3.1 Overall Approach

The project followed a systematic data science pipeline: *data acquisition* → *data cleaning* → *feature engineering* → *exploratory analysis* → *visualisation* → *interactive dashboard deployment*. Each phase is described in detail below.

3.2 Milestone 1 — Data Preparation

3.2.1 Environment Setup

The project was developed in Google Colab using the following core Python libraries:

Table 3.1: Python Libraries Used

Library	Purpose
pandas	Data loading, cleaning, aggregation, and transformation
numpy	Numerical operations and array handling
matplotlib	Static charting (histograms, line charts, bar charts)
seaborn	Statistical visualisation (heatmaps, box plots, scatter plots)
plotly	Interactive charts and choropleth maps
streamlit	Web-based interactive dashboard framework

3.2.2 Dataset Inspection

Upon loading the raw CSV, the following inspection steps were performed:

```
1 # Load dataset
2 df = pd.read_csv("GlobalWeatherRepository.csv")
3
4 # Inspect shape and types
```

```
5 df.info()
6 print("Shape:", df.shape)
7 print(df.dtypes)
8
9 # Statistical summary
10 df.describe()
```

Listing 3.1: Dataset Inspection Code

The inspection revealed:

- Duplicate columns in both metric (SI) and imperial units
- Timestamp stored as a raw string in the `last_updated` field
- Several columns with sparse information (text descriptions, epoch timestamps)
- Missing values in numeric columns requiring imputation

3.2.3 Missing Value Handling

Missing values were identified and handled using a two-pass imputation strategy:

```
1 # Count missing values
2 missing = df.isnull().sum().sort_values(ascending=False)
3 missing_percent = (df.isnull().sum() / len(df)) * 100
4
5 # Numeric columns: fill with median (robust to outliers)
6 numeric_cols = df.select_dtypes(include=np.number).columns
7 df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].
8     median())
9
10 # Categorical columns: fill with mode
11 categorical_cols = df.select_dtypes(include='object').columns
12 for col in categorical_cols:
13     df[col].fillna(df[col].mode()[0], inplace=True)
```

Listing 3.2: Missing Value Imputation

Rationale: The *median* was chosen over the *mean* for numeric imputation because weather variables such as precipitation are highly right-skewed — the median is resistant to extreme outliers and better represents the central tendency in such distributions.

3.2.4 Column Removal

Redundant and low-utility columns were dropped to reduce dimensionality:

```
1 columns_to_drop = [  
2     'temperature_fahrenheit',    # Redundant: Celsius retained  
3     'wind_mph',                  # Redundant: m/s computed  
4     'pressure_in',               # Redundant: Pascal computed  
5     'precip_in',                 # Redundant: mm retained  
6     'feels_like_fahrenheit',  
7     'visibility_miles',  
8     'gust_mph',  
9     'air_quality_gb-defra-index',  
10    'last_updated_epoch',        # Redundant: datetime parsed  
11    ]  
12 df.drop(columns=columns_to_drop, inplace=True)
```

Listing 3.3: Redundant Column Removal

3.2.5 Unit Conversion & Feature Engineering

All measurements were standardised to SI units:

```
1 # Wind speed: km/h --> m/s  
2 df['wind_mps'] = df['wind_kph'] * (1000 / 3600)  
3  
4 # Atmospheric pressure: mb --> Pascal  
5 df['pressure_pa'] = df['pressure_mb'] * 100  
6  
7 # Visibility: km --> metres  
8 df['visibility_m'] = df['visibility_km'] * 1000
```

```

9
10 # Gust speed: km/h --> m/s
11 df['gust_mps'] = df['gust_kph'] * (1000 / 3600)

```

Listing 3.4: Unit Conversions to SI

Unit Conversion Formulas Used:

$$v_{m/s} = v_{km/h} \times \frac{1000}{3600} \approx v_{km/h} \times 0.27 \quad (3.1)$$

$$P_{Pa} = P_{mb} \times 100 \quad (3.2)$$

$$d_m = d_{km} \times 1000 \quad (3.3)$$

3.2.6 Datetime Parsing and Temporal Features

The timestamp column was parsed and decomposed into year, month, and day components to enable temporal aggregations:

```

1 df['last_updated'] = pd.to_datetime(df['last_updated'])
2 df['year'] = df['last_updated'].dt.year
3 df['month'] = df['last_updated'].dt.month
4 df['day'] = df['last_updated'].dt.day

```

Listing 3.5: Datetime Parsing

3.2.7 Monthly Aggregation

Daily records were aggregated to monthly averages per country and location. This reduces noise from day-to-day variability and provides a manageable time series for trend analysis:

```

1 monthly_data = df.groupby(
2     ['country', 'location_name', 'year', 'month']
3 ).agg({
4     'temperature_celsius' : 'mean',
5     'wind_mps'            : 'mean',
6     'pressure_pa'         : 'mean',
7     'precip_mm'           : 'mean',
8     'humidity'            : 'mean',

```

```
9      'cloud' : 'mean',
10     'uv_index' : 'mean',
11     'air_quality_PM2.5' : 'mean',
12     'air_quality_PM10' : 'mean',
13 }) .reset_index()
```

Listing 3.6: Monthly Aggregation

Output Files:

- `cleaned_global_weather_daily.csv` — full cleaned daily dataset
- `cleaned_global_weather_monthly.csv` — monthly aggregated dataset (used by the dashboard)

3.3 Milestone 2 — Core Analysis & Visualisation Design

3.3.1 Statistical Analysis

Distribution Analysis

Histograms were plotted for key variables to understand their distributional shape:

```
1 plt.figure(figsize=(8, 5))
2 plt.hist(df['temperature_celsius'], bins=30,
3         color='tomato', edgecolor='white')
4 plt.title('Temperature Distribution')
5 plt.xlabel('Temperature ( C )')
6 plt.ylabel('Count')
7 plt.show()
```

Listing 3.7: Distribution Plots

Key observations from distributions:

- **Temperature** follows a broad, roughly bimodal distribution reflecting the diversity of climate zones (tropical vs. polar regions).
- **Humidity** is left-skewed with most records concentrated between 60% and 90%.

Correlation Analysis

A Pearson correlation matrix was computed across all numerical weather variables to identify linear relationships:

```
1 num_cols = ['temperature_celsius', 'wind_mps', 'pressure_pa',
2             'precip_mm', 'humidity', 'cloud', 'uv_index']
3 sns.heatmap(df[num_cols].corr(), annot=True, fmt='.2f',
4             cmap='coolwarm', center=0)
```

Listing 3.8: Correlation Heatmap

Pearson Correlation Coefficient:

$$r_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (3.4)$$

where $r_{XY} \in [-1, 1]$: values near +1 indicate strong positive correlation, near -1 strong negative correlation, and near 0 no linear relationship.

Seasonal and Yearly Trend Analysis

Monthly and yearly averages were computed to reveal seasonal cycles and multi-year temperature trends:

```
1 # Monthly average temperature (seasonal pattern)
2 monthly_temp = df.groupby('month')['temperature_celsius'].
   mean()
3
4 # Yearly average temperature (long-term trend)
5 yearly_temp = df.groupby('year')['temperature_celsius'].mean
   ()
```

Listing 3.9: Seasonal and Trend Analysis

Regional Comparison

The top 10 hottest and coldest countries were identified by ranking countries on their mean annual temperature:

```
1 top_hot = df.groupby('country')['temperature_celsius'] \
```

```

2         .mean().sort_values(ascending=False).head(10)
3
4 top_cold = df.groupby('country')['temperature_celsius'] \
5         .mean().sort_values(ascending=True).head(10)

```

Listing 3.10: Regional Ranking

3.3.2 Extreme Weather Event Detection

Extreme weather events were flagged using threshold-based rules derived from meteorological standards:

```

1 # Thresholds
2 T_HEAT = 40.0      # Heatwave: temperature > 40 C
3 T_RAIN = 25.0      # Heavy rain: precipitation > 25 mm (WMO)
4 T_WIND = 16.67     # High wind: > 16.67 m/s (~60 km/h)
5
6 df['is_heatwave'] = (df['temperature_celsius'] > T_HEAT).
7     astype(int)
8 df['is_heavy_rain'] = (df['precip_mm'] > T_RAIN).astype(int)
9 df['is_high_wind'] = (df['wind_mps'] > T_WIND).astype(int)

```

Listing 3.11: Extreme Event Flagging

Table 3.2: Extreme Weather Thresholds

Event Type	Threshold	Standard / Source
Heatwave	$T > 40^{\circ}\text{C}$	Global climatology benchmark
Heavy Rain	$P > 25\text{ mm}$	WMO daily heavy-rain boundary
High Wind	$v > 16.67\text{ m/s}$	Beaufort Scale 7 ($\approx 60\text{ km/h}$)
Cold Stress	$T < 0^{\circ}\text{C}$	Freezing point threshold
High UV	UV Index > 8	WHO Very High UV exposure level
Poor Air (PM2.5)	$> 25\text{ }\mu\text{g/m}^3$	WHO AQI moderate threshold

3.3.3 Climate Risk Score Formula

A composite Climate Risk Score was engineered to rank countries by multi-hazard exposure:

Climate Risk Score (0–100):

$$\begin{aligned}
\text{Risk} = & 30 \cdot \mathbf{1}[T > 35] + 15 \cdot \mathbf{1}[30 < T \leq 35] + 25 \cdot \mathbf{1}[P > 10] \\
& + 15 \cdot \mathbf{1}[P < 0.5] + 20 \cdot \mathbf{1}[v > 15] + 15 \cdot \mathbf{1}[\text{UV} > 8] \\
& + 10 \cdot \mathbf{1}[\text{PM}_{2.5} > 25] + 20 \cdot \mathbf{1}[T < 0]
\end{aligned} \tag{3.5}$$

clipped to $[0, 100]$, where $\mathbf{1}[\cdot]$ is the indicator function, T is temperature in $^{\circ}\text{C}$, P is precipitation in mm, and v is wind speed in m/s.

3.3.4 Year-over-Year Temperature Anomaly

The temperature anomaly quantifies how much the current year's average deviates from the previous year:

Year-over-Year Anomaly:

$$\Delta T = \bar{T}_{\text{year}} - \bar{T}_{\text{year}-1} \tag{3.6}$$

where $\bar{T}_{\text{year}}$ is the mean temperature for the selected year and $\bar{T}_{\text{year}-1}$ is the mean for the prior year. A positive anomaly indicates warming; negative indicates cooling.

3.3.5 Visualisation Type Selection

Table 3.3: Visualisation Types Selected and Their Justification

Chart Type	Variables	Justification
Choropleth Map	Temperature, Precipitation, Risk Score	Geographic comparison across countries
Violin + Box Plot	Temperature by month	Shows distribution shape, spread, and outliers
Line Chart	Monthly trends (all variables)	Reveals seasonal cycles and temporal trends
Correlation Heatmap	All numeric variables	Identifies linear relationships
Scatter Plot	Temperature vs Humidity	Reveals bivariate relationship patterns
Treemap	Top-N countries by temperature/risk	Proportional comparison of many entities
Radar / Spider Chart	Multi-metric country comparison	Simultaneous comparison on multiple axes
Polar Bar/Line	Seasonal wind, cloud cover	Circular representation of monthly cycles
Waterfall Chart	Month-over-month temperature change	Cumulative sequential changes
Gauge Indicator	Risk score, UV index	Single-value status indicator
Dumbbell Chart	Monthly temp comparison: two countries	Highlights range and gap between two series
Funnel Chart	Top 8 hottest countries	Ordered ranking with visual hierarchy

3.4 Milestone 3 — Dashboard Development

3.4.1 Technology Stack

Table 3.4: Dashboard Technology Stack

Component	Role
Streamlit	Web application framework; layout, sidebar, tabs
Plotly Express	High-level interactive chart generation
Plotly Graph Objects	Low-level chart customisation (subplots, shapes)
pandas	Data filtering and real-time aggregation
numpy	Numerical operations (normalisation, clipping)
CSS	Custom styling (KPI cards, tab animations, colour themes)
JavaScript	Tab navigation arrows injected into the Streamlit DOM

3.4.2 Dashboard Architecture

The dashboard was structured into **8 thematic tabs**, each addressing a distinct analytical dimension:

- Tab 1. Executive Overview** — KPI cards (avg temperature, anomaly, precipitation, wind, extreme events, hottest country), climate fingerprint heatmap, treemap, normalised multi-metric trend, humidity/cloud/UV overlay, scatter plot, dual-axis bar+line chart.
- Tab 2. Temperature Intelligence** — Violin plots, choropleth map, waterfall chart, funnel chart, lollipop chart (coldest countries), range bar chart.
- Tab 3. Precipitation & Wind** — Stacked area chart, wind polar chart, horizontal bar chart, grouped bar chart, global precipitation choropleth.
- Tab 4. Extreme Events** — Gauge indicator, extreme event bar chart, event-type heatmap, multi-line event timeline.

Tab 5. Regional Comparison — Side-by-side country selector, radar chart, dumbbell chart, wind/humidity/UV/precipitation/cloud comparison charts.

Tab 6. Humidity & Cloud — Monthly trend line, polar bar chart (cloud cover), lollipop chart, grouped bar chart, dual overlay chart.

Tab 7. UV & Air Quality — UV gauge + trend subplot, PM2.5 diverging bar chart, PM2.5/PM10 area chart, dual-axis UV+PM2.5 timeline, ranked bar charts.

Tab 8. Climate Risk Index — Risk gauge, monthly risk timeline, risk treemap, top-15 risk bar chart, global risk choropleth.

3.4.3 Sidebar Filters

Two global filters were provided in the sidebar to allow real-time data slicing:

- **Country Selector:** Single-select dropdown; options include “All Countries” (global view) and each individual country in the dataset.
- **Year Selector:** Single-select dropdown of all available years.

All charts, KPI cards, and maps update dynamically when a filter is changed, without requiring a page reload, leveraging Streamlit’s reactive execution model.

3.4.4 KPI Cards

Each tab displays 4–6 KPI (Key Performance Indicator) cards rendered as styled HTML components. Each card shows a metric label, primary value, and a contextual subtitle. Cards use a dark background with a coloured top border corresponding to the metric theme (e.g., red for temperature, blue for precipitation).

3.4.5 Performance Optimisations

The dashboard uses `@st.cache_data` decorators to memoize expensive aggregation functions. This ensures that repeated filter interactions do not re-execute full `groupby` operations on the entire dataset:

```
1 @st.cache_data
2 def compute_monthly_agg(df, country, year):
3     """Cached monthly aggregation per country/year."""
4     ...
```

```
5
6 @st.cache_data
7 def compute_risk_scores(df, year):
8     """Cached vectorised risk score computation."""
9     ...
```

Listing 3.12: Caching Strategy

4. Visualisations and Insights

This chapter presents the key visualisations developed across all three milestones, grouped by analytical theme. Dashboard screenshots should be inserted in the placeholder regions.

4.1 Executive Overview

Tab 1 — Executive Overview: This tab serves as the high-level entry point of the dashboard, providing a concise snapshot of the global or country-level climate state for the selected year. It presents six KPI cards covering average temperature, year-over-year anomaly, precipitation, wind speed, extreme event count, and the hottest country. Supporting visualisations include a climate fingerprint heatmap (normalised across six variables for the top 20 countries), a top-30 temperature treemap, a normalised multi-metric trend line, a humidity/cloud/UV overlay chart, a temperature-vs-humidity scatter plot, and a dual-axis bar and line chart. Together these elements give decision-makers an immediate, comprehensive overview without requiring deeper tab navigation.

4.1.1 KPI Summary Cards

The top row of the Executive Overview tab presents six KPI cards at a glance: average temperature, year-over-year anomaly, precipitation, wind speed, number of extreme events, and the hottest country. These cards provide an immediate snapshot of the climate state for the selected scope and year.

Key Design Decisions:

- For *All Countries* mode, precipitation is shown as the *average monthly* value (meaningful for global comparison).
- For a *single country*, precipitation is shown as the *annual total* (more relevant for water availability analysis).

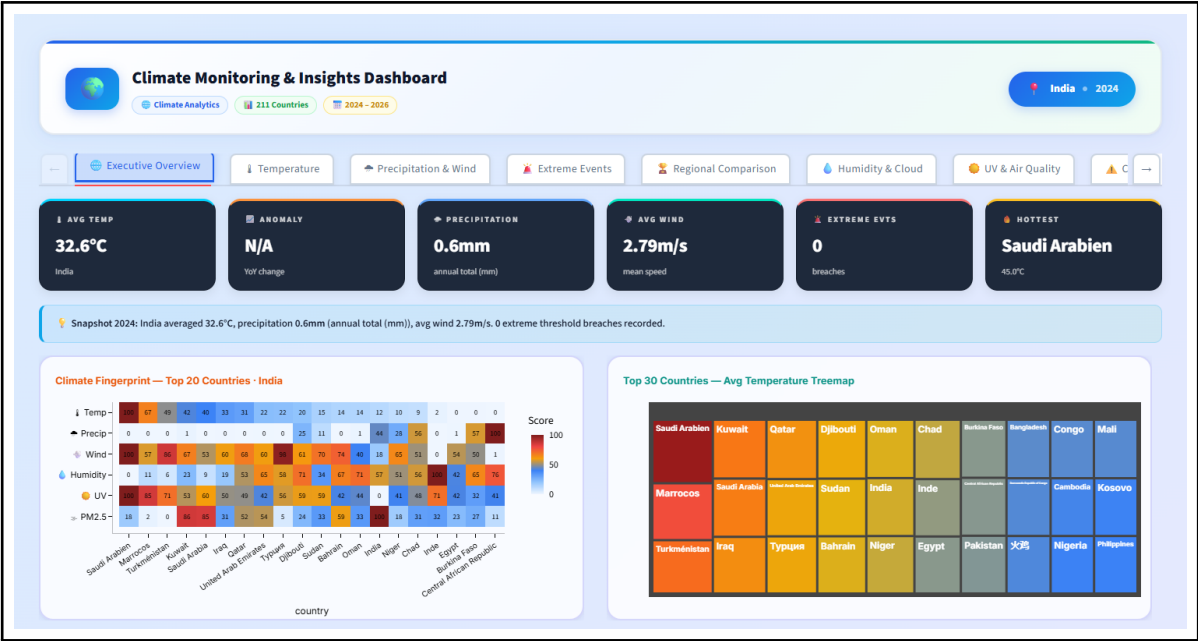


Figure 4.1: Executive Overview — KPI Cards and Charts

4.1.2 Climate Fingerprint Heatmap

The climate fingerprint heatmap normalises six variables (Temperature, Precipitation, Wind, Humidity, UV, PM2.5) to a 0–100 scale for the top 20 countries. This allows visual comparison of relative climate intensities across countries on a unified scale.

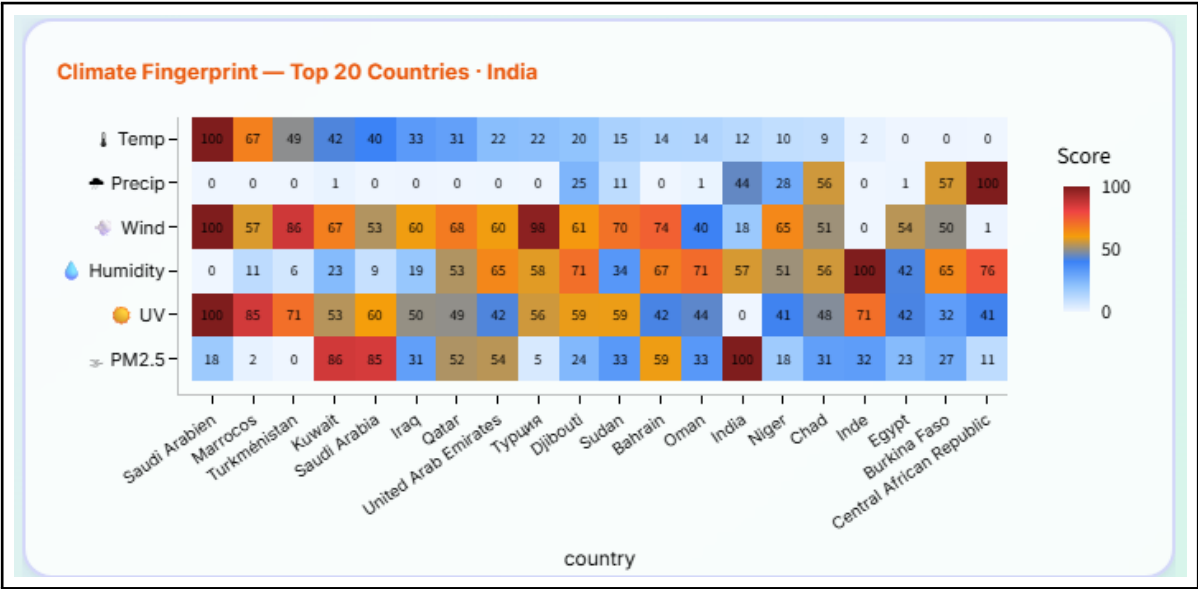


Figure 4.2: Climate Fingerprint Heatmap — Top 20 Countries

Normalisation Formula:

$$\hat{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \times 100 \tag{4.1}$$

4.1.3 Top-30 Countries Temperature Treemap

A treemap displays the top 30 countries ranked by average temperature. Cell area is proportional to average temperature value; colour intensity follows a blue-to-red scale to intuitively convey coldness versus heat.

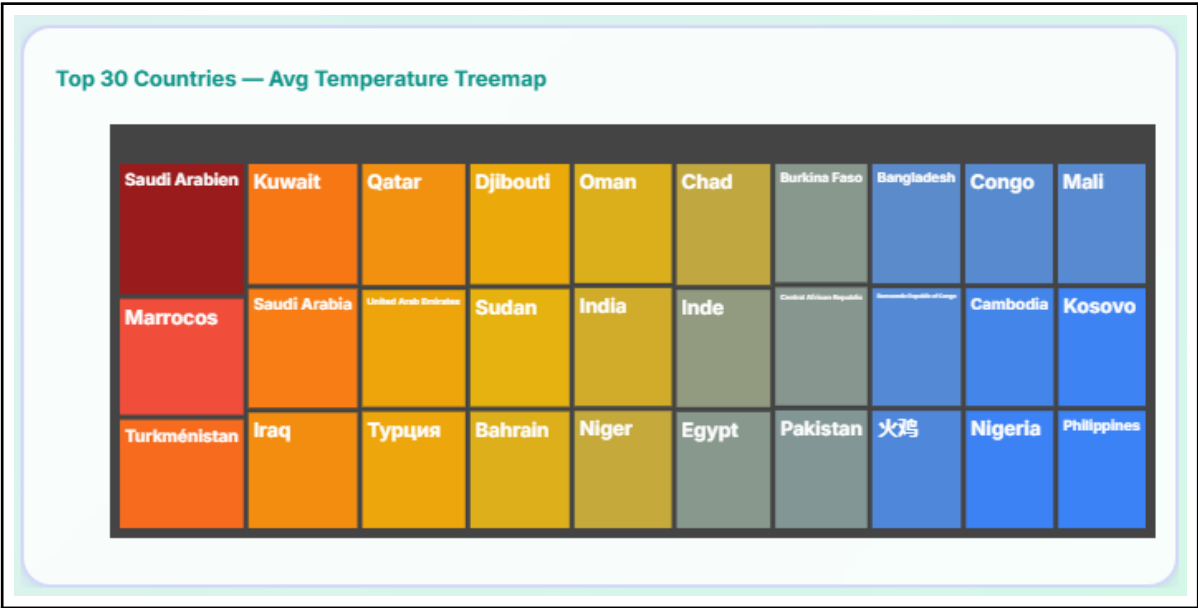


Figure 4.3: Top-30 Countries Temperature Treemap

4.1.4 Normalised Multi-Metric Monthly Trend

A multi-line chart normalises six climate variables — Temperature, Humidity, UV Index, Wind Speed, Precipitation, and PM2.5 — to a common 0–100 scale and plots them across all twelve months as overlapping lines. Each variable is assigned a distinct colour, allowing the viewer to identify which metrics peak together and which are inversely related. Hover tooltips display the actual (un-normalised) values, preserving analytical precision while the visual comparison is made on a shared axis.

Design Note: Normalisation is performed independently for each variable so that a metric with a small absolute range (e.g., UV Index 0–12) is visually comparable to one with a large range (e.g., Temperature –30 to 50 °C). This prevents high-

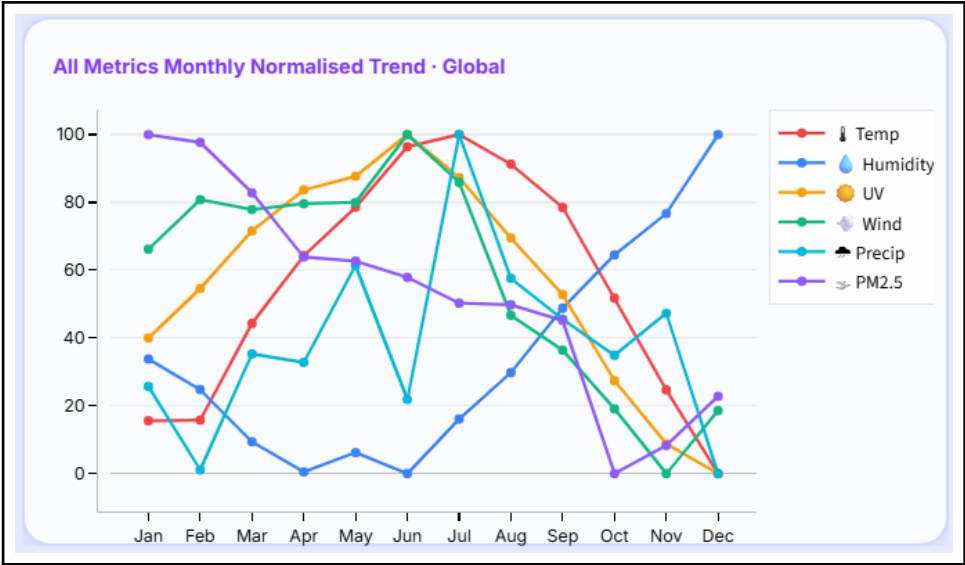


Figure 4.4: All Metrics Monthly Normalised Trend — Executive Overview

magnitude variables from dominating the chart scale.

4.1.5 Monthly Humidity, Cloud Cover, and UV Overlay

A stacked fill-area chart overlays three atmospheric variables — Humidity (%), Cloud Cover (%), and UV Index ($\times 10$ for scale alignment) — on a shared monthly axis. The humidity area fills to zero, the cloud cover area fills from the humidity trace upwards (tonexty), and the UV index is rendered as a dotted line with circular markers. This layered representation reveals the co-variation between moisture, cloud formation, and solar radiation intensity across the year.

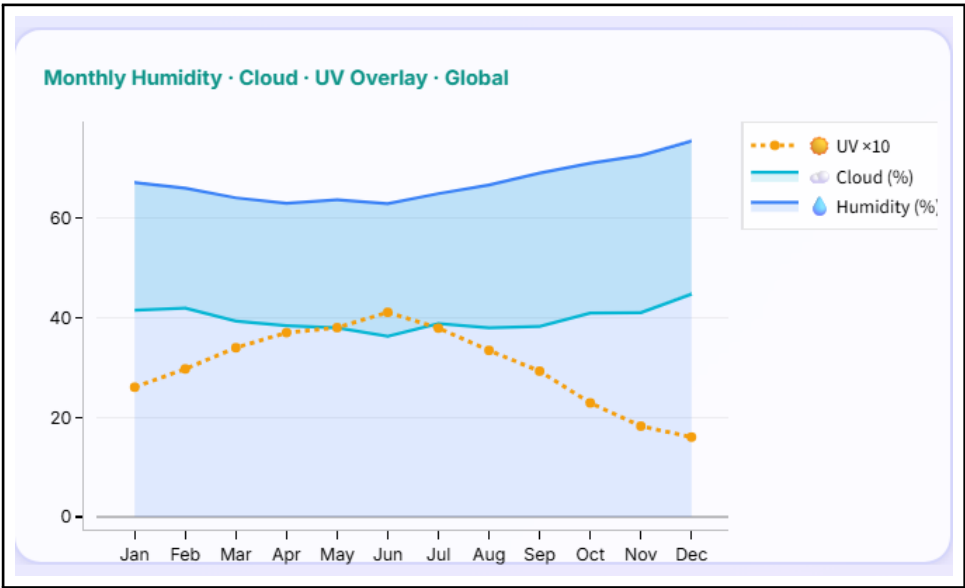


Figure 4.5: Monthly Humidity, Cloud Cover, and UV Overlay — Executive Overview

4.1.6 Humidity vs Temperature Scatter Plot

A scatter plot maps average humidity against average temperature for every country in the dataset. Each point represents one country; colour encodes average precipitation intensity using a yellow-to-blue colour scale, and marker size is uniform to avoid overplotting. This bivariate chart reveals that higher temperatures are generally associated with lower humidity in arid regions, while equatorial countries simultaneously exhibit high temperature and high humidity. Country name tooltips are shown on hover.

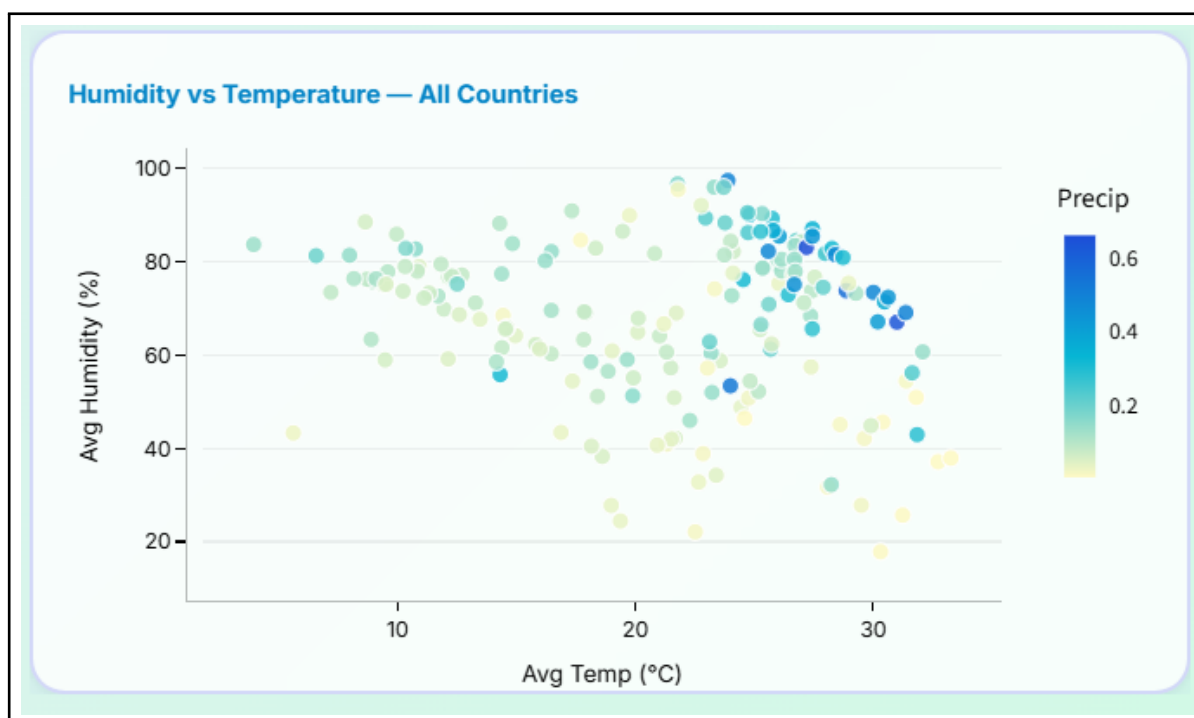


Figure 4.6: Humidity vs Temperature Scatter Plot — All Countries

4.1.7 Monthly Temperature and Precipitation — Dual-Axis Bar and Line Chart

A dual-axis combination chart renders monthly average temperature as vertical bars on the primary y -axis and monthly average precipitation as a line with circular markers on the secondary y -axis. This layout allows the viewer to immediately compare temperature peaks and precipitation cycles in the same panel, making seasonal relationships — such as high-rainfall monsoon months coinciding with moderate temperature — visually apparent.

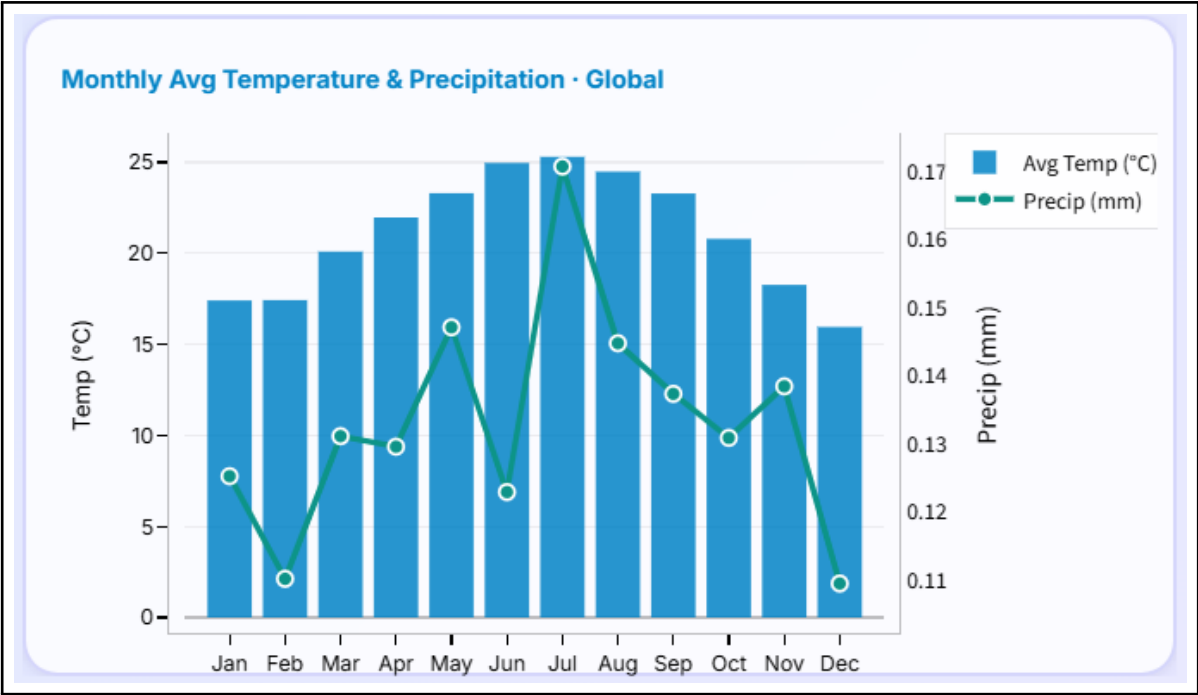


Figure 4.7: Monthly Average Temperature and Precipitation — Dual-Axis Chart

4.2 Temperature Intelligence

Tab 2 — Temperature Intelligence: This tab provides an in-depth exploration of temperature patterns at both global and country-specific levels. It opens with a monthly violin plot that reveals the full distributional shape of daily temperatures for each month, with an embedded box plot for quartile reference and an optional country overlay. A global choropleth map colours every country by its mean annual temperature on a diverging cold-to-hot scale. Additional charts include a month-over-month waterfall chart highlighting sequential temperature changes, a funnel chart of the top 8 hottest countries, a lollipop chart ranking the 10 coldest countries, and a range bar chart.

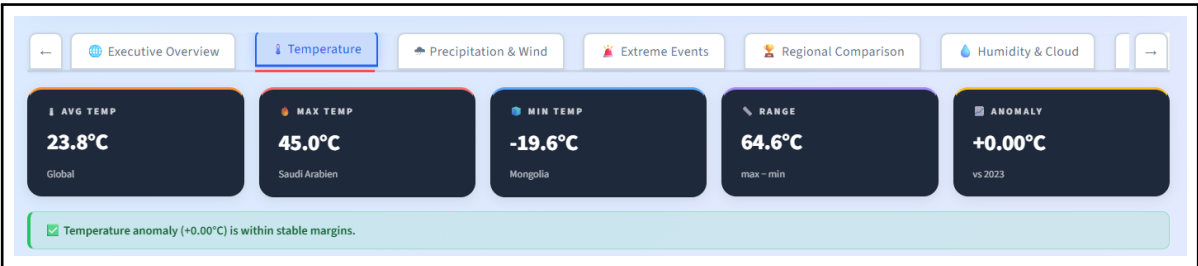


Figure 4.8: Temperature tab KPI cards

4.2.1 Monthly Temperature Violin Plot

The violin plot shows the full distribution of daily temperatures for every month across all countries. A box plot is embedded inside the violin for quartile reference. When a specific country is selected, diamond markers overlay the global violins with that country's monthly average, enabling quick comparison against global norms.

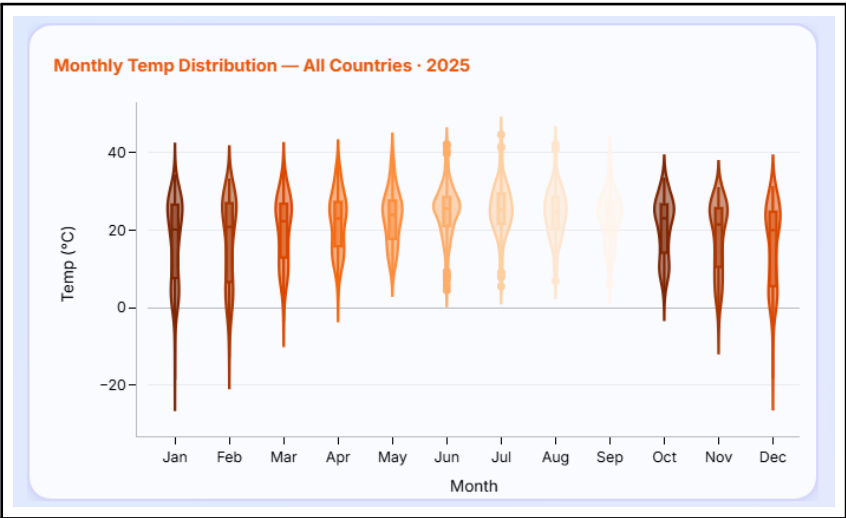


Figure 4.9: Monthly Temperature Violin Plot (Global Distribution with Country Overlay)

4.2.2 Global Temperature Choropleth Map

The world choropleth map colours each country by its mean annual temperature, using a diverging colour scale from deep blue (cold) to dark red (hot). A highlighted gold border is drawn on the currently selected country.

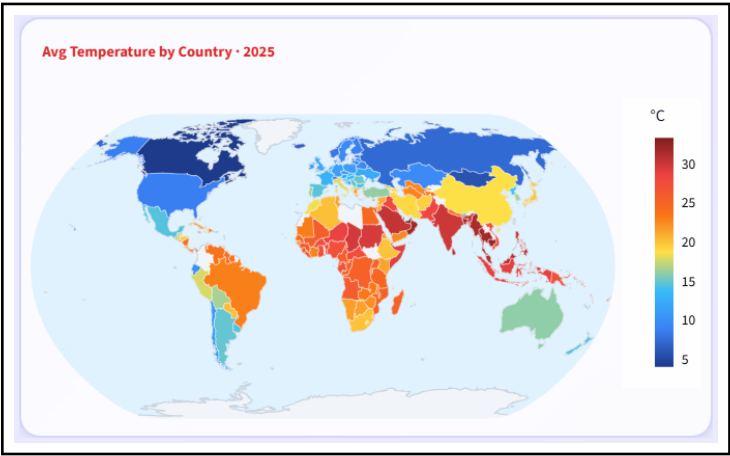


Figure 4.10: Global Average Temperature Choropleth Map

4.2.3 Month-over-Month Waterfall Chart

The waterfall chart shows the sequential temperature change from month to month. Red bars indicate months warmer than the previous; green bars indicate cooler months. The first bar anchors to January’s absolute value; subsequent bars show relative change.

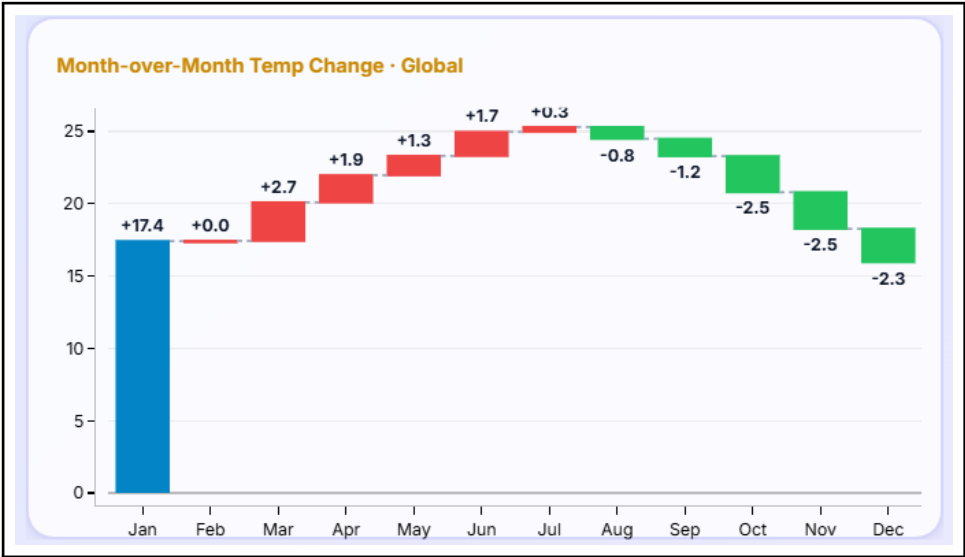


Figure 4.11: Month-over-Month Temperature Waterfall Chart

4.2.4 Top-10 Coldest Countries — Lollipop Chart

A lollipop (dot + line) chart ranks the 10 coldest countries. Labels are dynamically positioned to the left of the dot for negative temperatures and to the right for positive values, preventing label overlap with the zero axis.

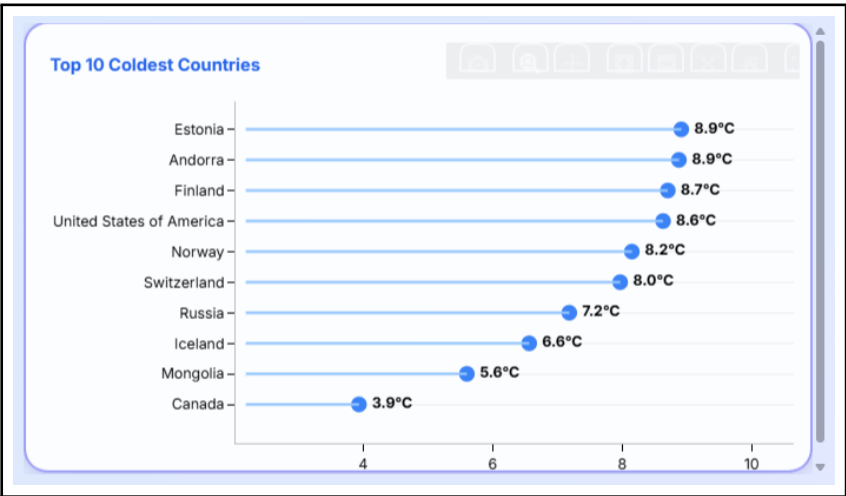


Figure 4.12: Top-10 Coldest Countries Lollipop Chart

4.2.5 Top-8 Hottest Countries — Funnel Chart

A funnel chart ranks the eight hottest countries in descending order of average temperature. Each tier of the funnel is filled with a shade from the Reds colour palette, reinforcing the heat metaphor through progressive colour intensification from the widest (hottest) band at the top to the narrowest at the bottom. Both the country name and average temperature value are displayed inside each funnel band for immediate readability.

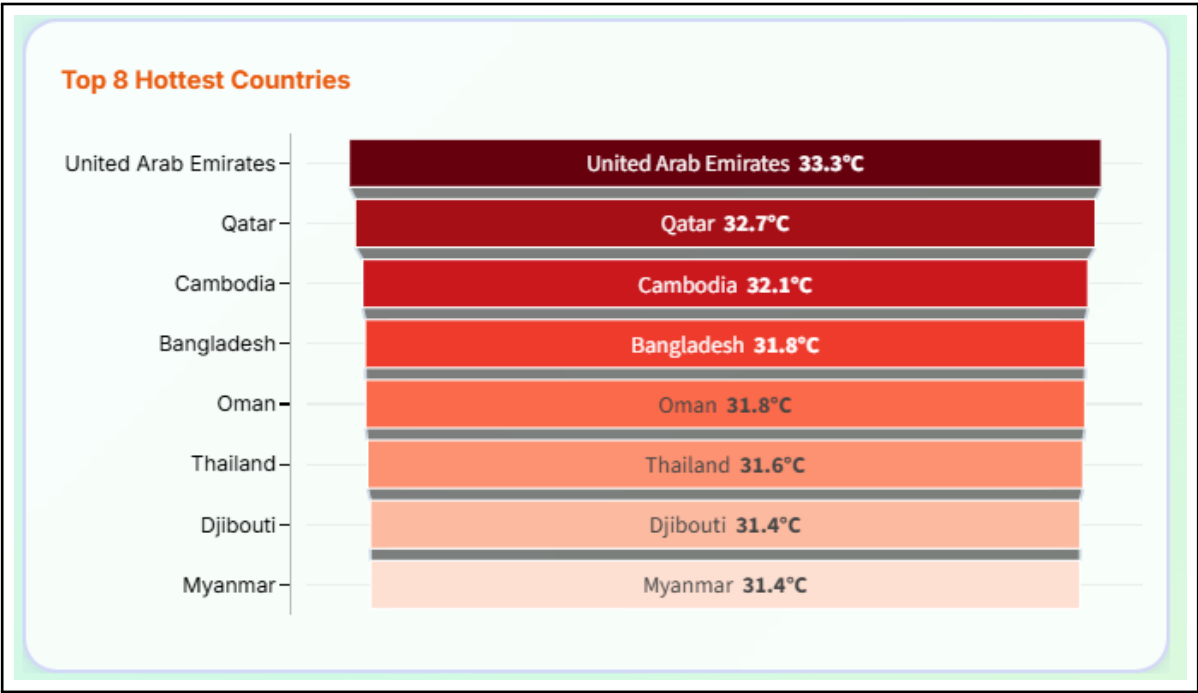


Figure 4.13: Top-8 Hottest Countries — Funnel Chart

4.2.6 Countries with Widest Temperature Range — Horizontal Bar Chart

A horizontal bar chart ranks the ten countries with the greatest intra-annual temperature range (maximum monthly average minus minimum monthly average). Bar colour encodes range magnitude using a purple gradient from light (narrow range) to deep indigo (extreme range). Value labels are shown outside each bar. This chart highlights countries in continental interior climates — such as Russia and Canada — that experience dramatic seasonal swings between winter cold and summer heat. I gives temperature difference so which gives anomaly temperature

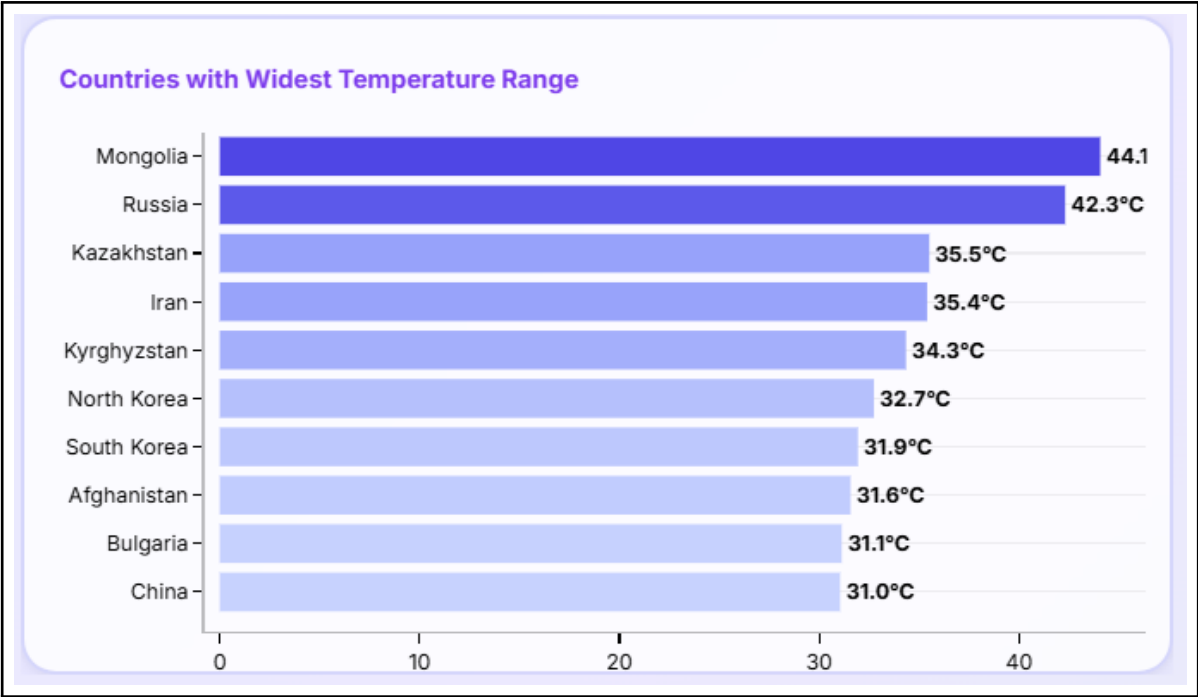


Figure 4.14: Countries with Widest Temperature Range — Horizontal Bar Chart

4.3 Precipitation & Wind

Tab 3 — Precipitation & Wind: This tab focuses on two of the most operationally significant weather variables — rainfall and wind. A stacked area chart overlays monthly average precipitation with rescaled humidity to expose the relationship between atmospheric moisture and rainfall. A polar (rose) line chart renders monthly wind speeds in a circular format, making seasonal wind cycles visually intuitive at a glance. A horizontal bar chart ranks the top 10 wettest countries by total annual precipitation, and a grouped bar chart enables multi-country precipitation comparison. A global precipitation choropleth closes the tab with a geographic perspective. This tab is particularly valuable for water resource planning and agricultural applications.

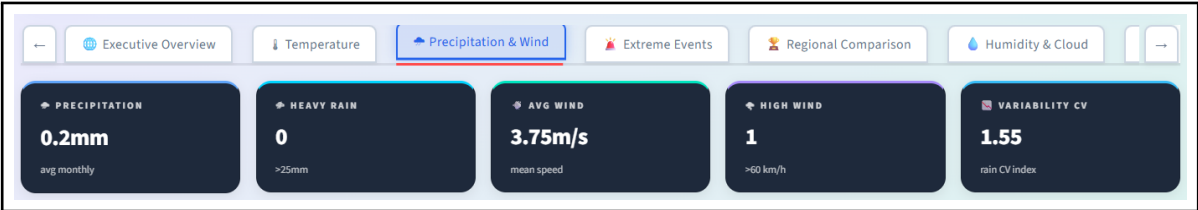


Figure 4.15: Precipitation & Wind Tab KPI cards

4.3.1 Monthly Precipitation & Humidity — Area Chart

A stacked area chart overlays monthly average precipitation (mm) with rescaled humidity ($\div 10$) to enable side-by-side visual comparison on a shared axis. The overlap reveals the relationship between atmospheric moisture and rainfall.

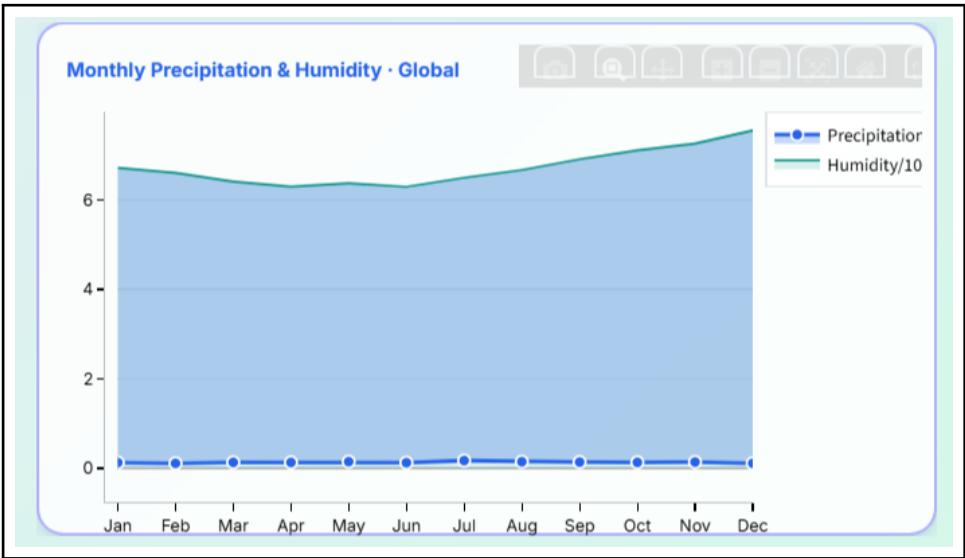


Figure 4.16: Monthly Precipitation and Humidity Stacked Area Chart

4.3.2 Wind Speed Polar Chart

A circular (polar) line chart plots monthly average wind speed across 12 months arranged around 360°. The closed polygon area makes seasonal wind patterns visually intuitive — a circle indicates uniform year-round wind; an elongated shape indicates strong seasonality.

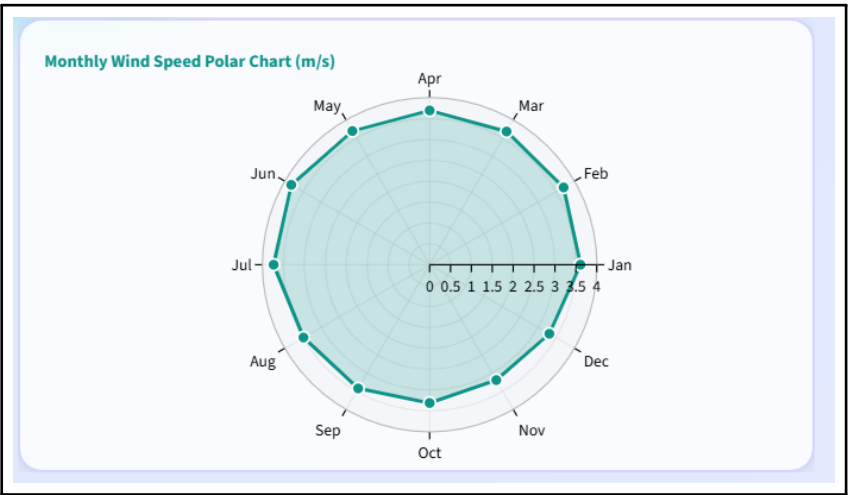


Figure 4.17: Monthly Wind Speed Polar Chart

4.3.3 Top-10 Countries by Total Precipitation

A horizontal bar chart ranks the 10 wettest countries by total annual precipitation. Gradient blue colouring reinforces the “water quantity” theme.

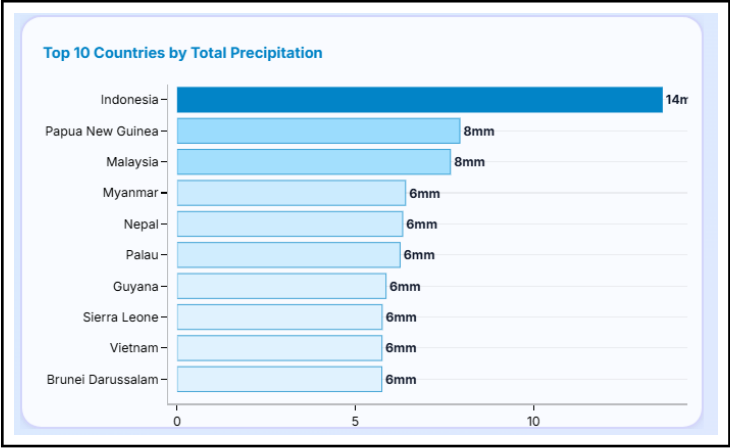


Figure 4.18: Top-10 Countries by Total Annual Precipitation

4.3.4 Monthly Precipitation and Wind Speed — Grouped Bar Chart

A grouped bar chart presents monthly average precipitation (mm) and wind speed (m/s) as paired bars for each month. Precipitation bars are rendered in blue and wind bars in teal, with both series sharing the same *y*-axis. This layout enables direct month-level comparison of two operationally important variables, immediately exposing months where both high rainfall and strong winds coincide — a critical factor in storm and flood risk assessment.

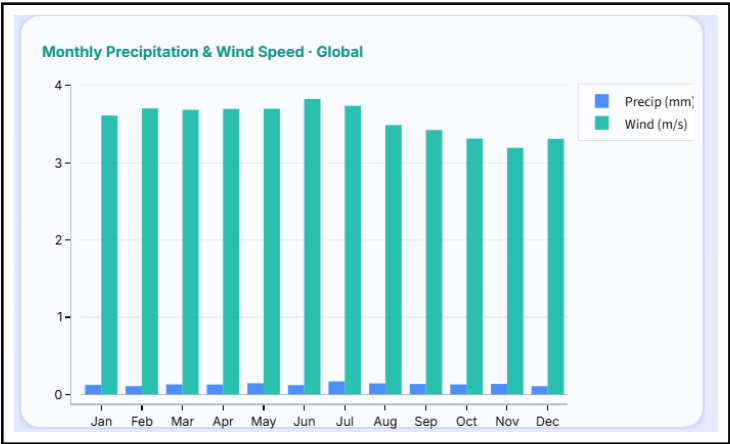


Figure 4.19: Monthly Precipitation and Wind Speed — Grouped Bar Chart

4.3.5 Global Average Precipitation Choropleth Map

A world choropleth map encodes each country’s mean monthly precipitation using a yellow-to-green colour scale, ranging from pale yellow (low precipitation) to dark forest green (high precipitation). This geographic representation immediately identifies the wettest equatorial and tropical regions — such as South and Southeast Asia and Central Africa — as well as arid desert zones in North Africa and the Middle East.

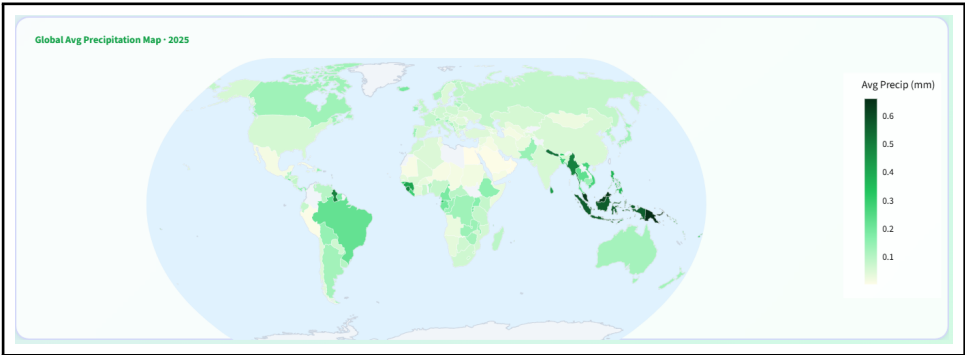


Figure 4.20: Global Average Precipitation Choropleth Map

4.4 Extreme Events

Tab 4 — Extreme Events: This tab is dedicated to identifying and quantifying dangerous weather conditions using threshold-based event detection. A gauge chart displays the composite Climate Risk Score on a 0–100 scale, with colour-coded Low/Medium/High zones and a delta indicator showing deviation from a baseline of 50. A bar chart summarises total extreme event counts by type. An annotated event-type-by-month heatmap reveals which months are most prone to each hazard category. A multi-series line chart then traces the monthly trajectory of each event type through the year using distinct colours and marker symbols. This tab is designed to support emergency preparedness, infrastructure planning, and climate risk reporting.

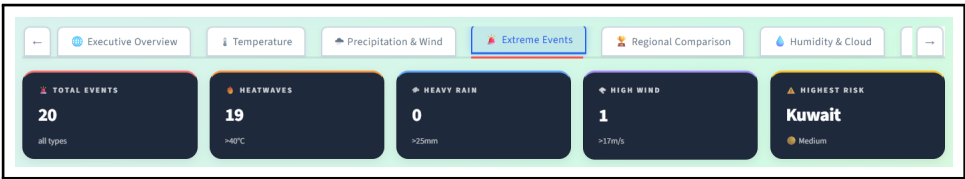


Figure 4.21: Extreme events KPI tab cards

4.4.1 Climate Risk Gauge

A gauge chart visualises the composite Climate Risk Score on a 0–100 scale. Three coloured bands distinguish Low (0–20), Medium (20–50), and High (50–100) risk zones. A delta indicator shows the change from a baseline of 50.

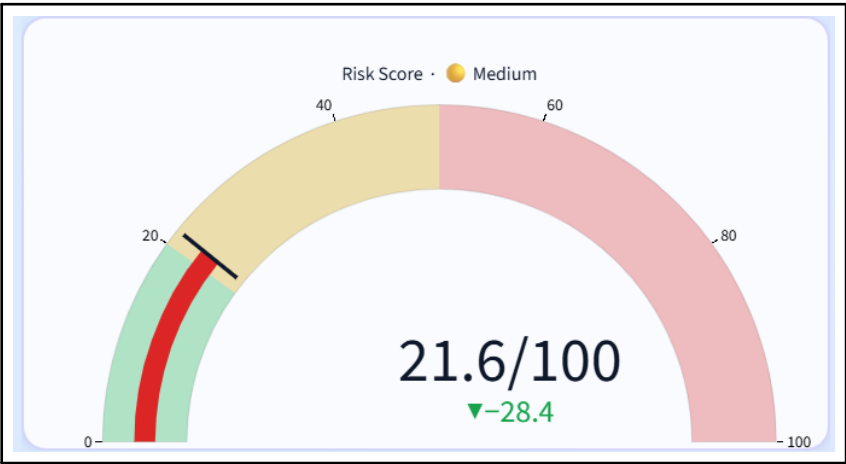


Figure 4.22: Climate Risk Score Gauge Indicator

4.4.2 Top-10 Countries by Extreme Events — Horizontal Bar Chart

A horizontal bar chart ranks the ten countries with the highest total count of threshold-breaching extreme events across all three categories (Heatwave, Heavy Rain, High Wind). Bar colour is conditionally assigned: green for fewer than three events, amber for three to five, and red for six or more, providing an immediate traffic-light-style severity signal. Value labels are placed outside each bar.

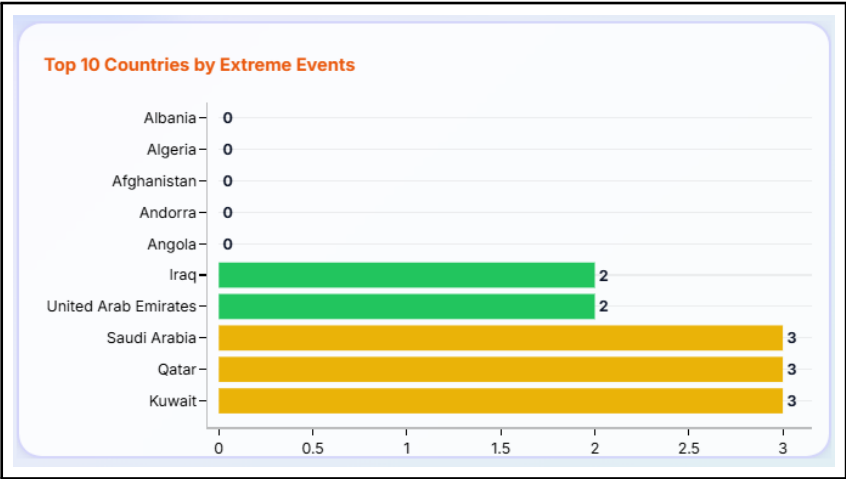


Figure 4.23: Top-10 Countries by Extreme Events — Horizontal Bar Chart

4.4.3 Extreme Events — Type × Month Heatmap

An annotated heatmap shows the count of extreme weather events (Heatwave, Heavy Rain, High Wind) across every month. Cell colour intensity reflects event frequency, immediately highlighting which months and event types are most severe.

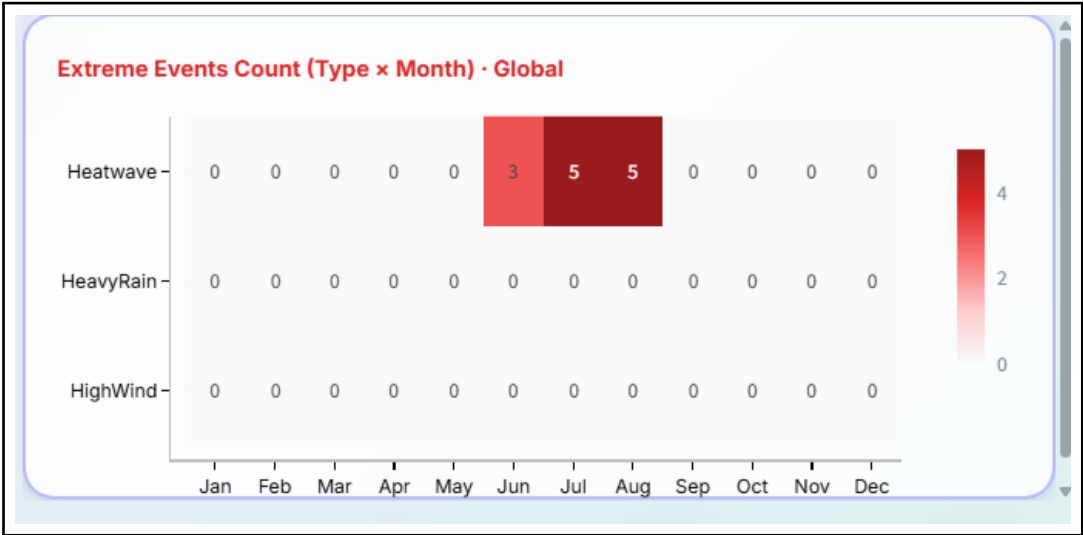


Figure 4.24: Extreme Events Type-by-Month Heatmap

4.4.4 Monthly Extreme Event Timeline

A multi-series line chart traces the monthly count of each extreme event type through the year, using distinct colours and marker symbols (circle, diamond, triangle) for easy differentiation.

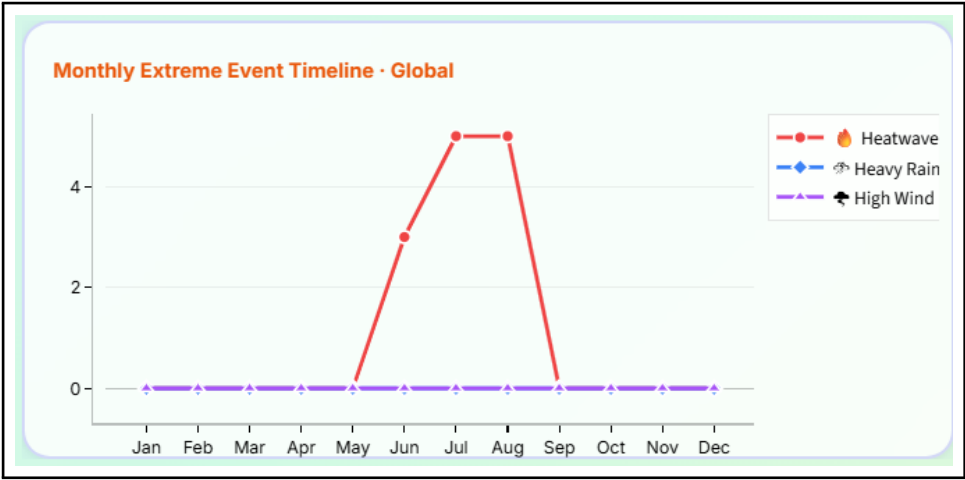


Figure 4.25: Monthly Extreme Event Timeline (Multi-Series Line Chart)

4.5 Regional Comparison

Tab 5 — Regional Comparison: This tab enables direct side-by-side analysis of two user-selected countries across the full suite of climate variables. A radar (spider) chart simultaneously compares both countries on six normalised metrics — Temperature, Precipitation, Wind, Humidity, UV, and PM2.5 — making their overall climate profiles immediately apparent. A dumbbell chart plots monthly average temperatures for each country as paired dots connected by a line, with line length indicating the temperature gap per month. A grouped bar chart presents all six metrics side-by-side for a single-view comparative summary. This tab is ideal for analysts comparing climate suitability between locations for agriculture, tourism, or infrastructure investment.

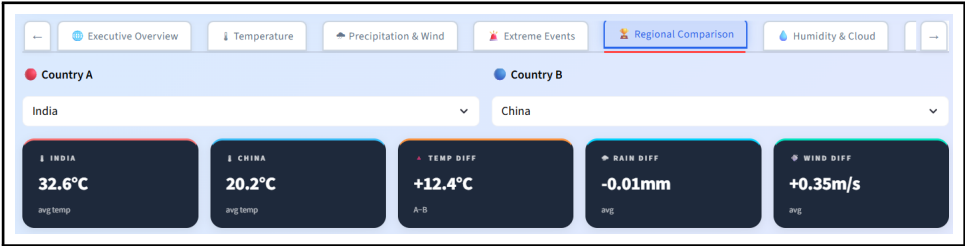


Figure 4.26: Regional Comparison tab KPI cards

4.5.1 Climate Radar Chart (Country A vs Country B)

A radar (spider) chart simultaneously compares two user-selected countries across six normalised metrics: Temperature, Precipitation, Wind, Humidity, UV, and PM2.5. The area of each polygon conveys overall climate intensity.

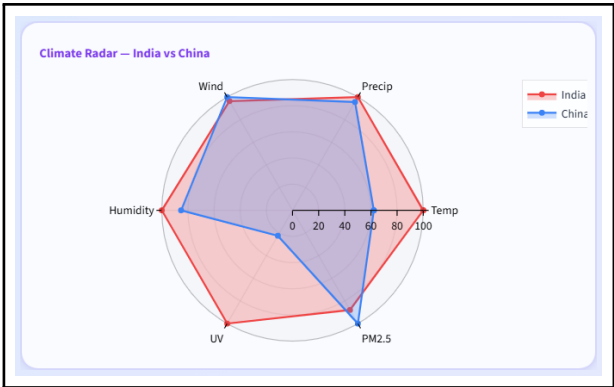


Figure 4.27: Climate Radar Chart — Country A vs Country B

4.5.2 Monthly Temperature Dumbbell Chart

For two selected countries, a dumbbell chart plots monthly average temperatures as two dots connected by a horizontal line. The length of each line segment indicates the temperature gap between the two countries in that month.

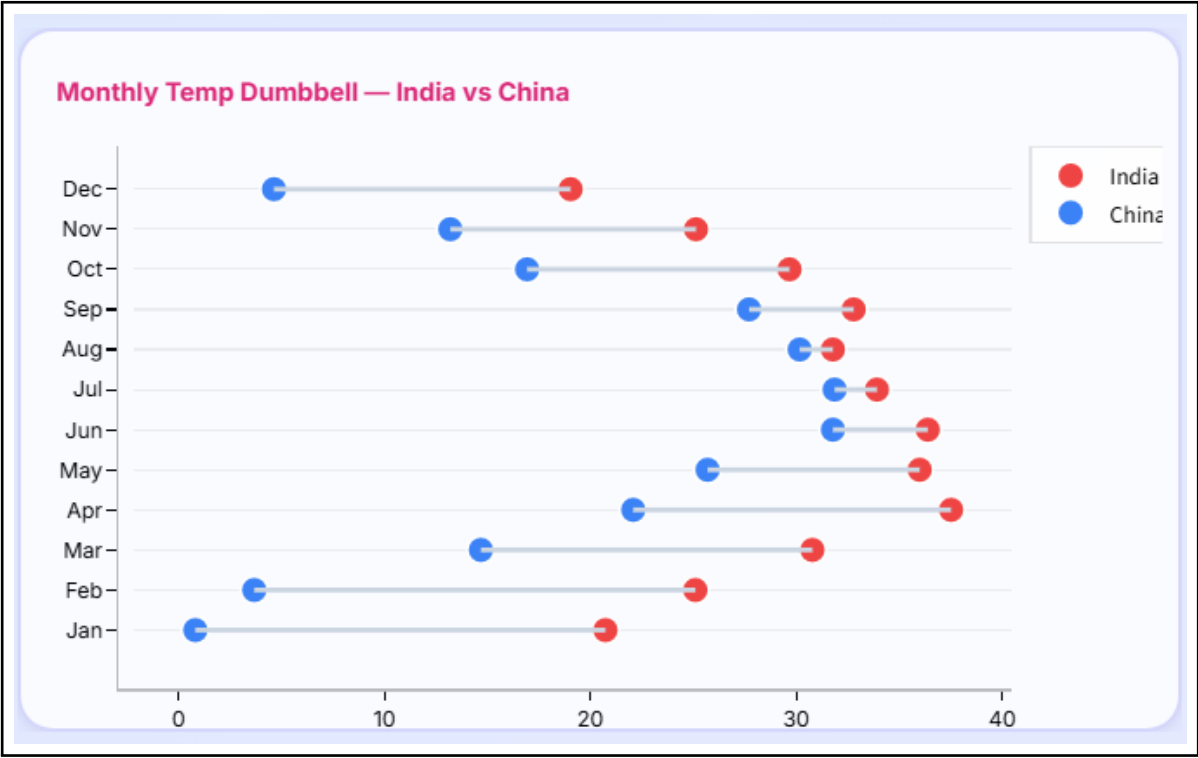


Figure 4.28: Monthly Temperature Dumbbell Chart

4.5.3 All Metrics Side-by-Side Bar Chart

A grouped bar chart presents all six climate metrics for both countries simultaneously. This provides a comprehensive single-view comparison.

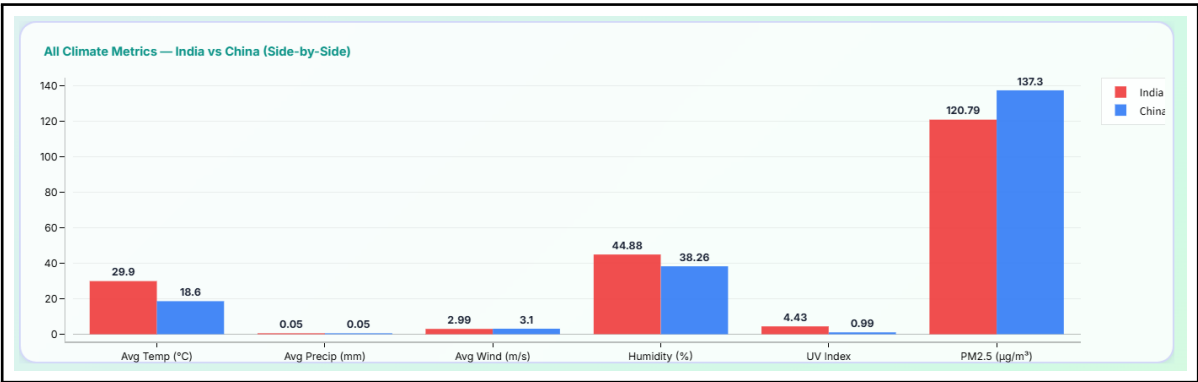


Figure 4.29: All Climate Metrics Side-by-Side Bar Chart

4.5.4 Monthly Wind Speed Comparison — Line Chart

A dual-series line chart plots monthly average wind speed for Country A (red) and Country B (blue) across all twelve months. Each series uses filled circular markers and smooth connecting lines. The chart allows viewers to identify months where one country experiences significantly stronger winds — for example, monsoon-driven gusts in South Asia peaking in mid-year — and months where wind behaviour converges.



Figure 4.30: Monthly Wind Speed Comparison — Country A vs Country B

4.5.5 Monthly Humidity Comparison — Line Chart

A dual-series line chart tracks monthly average humidity for the two selected countries. High humidity months associated with monsoon or tropical rainfall seasons are immediately visible as peaks in the respective country's trace. Cross-overs between the two lines reveal months when the humidity balance inverts between the two locations.

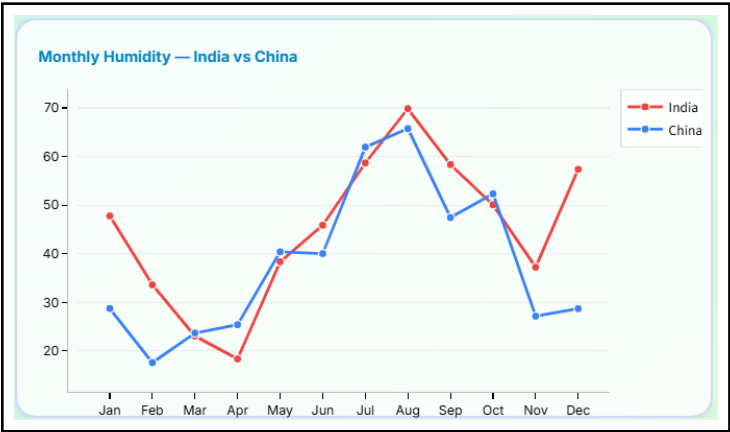


Figure 4.31: Monthly Humidity Comparison — Country A vs Country B

4.5.6 Monthly UV Index Comparison — Grouped Bar Chart

A grouped bar chart presents the monthly UV index for both countries as paired bars per month. A horizontal dashed red reference line marks UV level 8 — the WHO “Very High” exposure threshold — enabling immediate identification of months where either country exceeds safe UV limits. This chart is particularly relevant for agricultural, tourism, and public health planning.

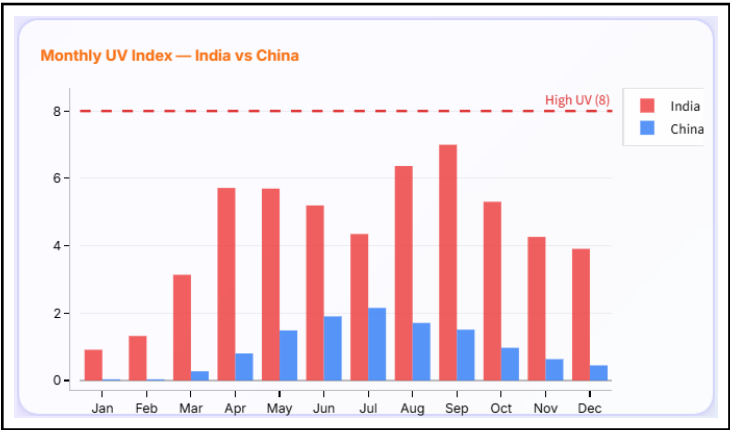


Figure 4.32: Monthly UV Index Comparison — Country A vs Country B

4.5.7 Monthly PM2.5 Comparison — Line Chart

A dual-series line chart compares monthly average PM2.5 air quality concentrations for the two selected countries. This visualisation directly supports environmental health comparisons between regions — for example, revealing whether one country’s air quality deteriorates during dry winter months while the other remains within safe bounds throughout the year.

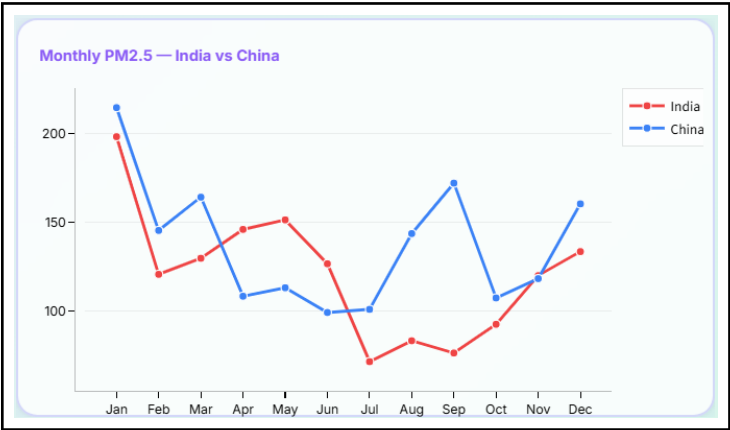


Figure 4.33: Monthly PM2.5 Comparison — Country A vs Country B

4.5.8 Average Cloud Cover Comparison — Donut Charts

Two side-by-side donut charts display the annual average cloud cover percentage for each selected country. The filled arc represents cloud cover and the remaining arc represents clear sky. A central annotation displays the numeric percentage value in the same colour as the country’s assigned trace colour. This layout provides an at-a-glance summary of overall sky clarity, useful for solar energy potential assessments.

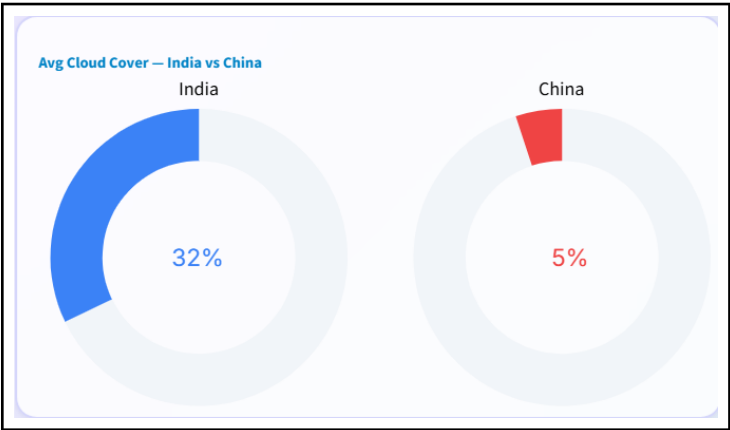


Figure 4.34: Average Cloud Cover Comparison — Donut Charts (Country A vs Country B)

4.5.9 Monthly Precipitation Comparison — Grouped Bar Chart

A grouped bar chart presents monthly average precipitation for both countries as paired bars per month. The side-by-side layout makes it straightforward to identify months where one country receives substantially more rainfall — such as a monsoon peak — and months where both countries experience similar precipitation levels.



Figure 4.35: Monthly Precipitation Comparison — Country A vs Country B

4.6 Humidity & Cloud Cover

Tab 6 — Humidity & Cloud Cover: This tab explores two closely related atmospheric variables that influence perceived weather comfort and solar energy potential. A smooth spline area chart plots the monthly humidity trend with a horizontal dashed line marking the annual mean, making it easy to identify months that are wetter or drier than average. A polar bar (rose) chart arranges cloud cover values for all 12 months around a circle, with bar length representing average cloud cover and colour transitioning from light blue (clear) to dark navy (overcast). A lollipop chart ranks countries or months by humidity, and a grouped bar chart allows multi-country comparison. A dual overlay chart combines humidity and cloud cover on a shared axis to reveal their co-variation. This tab supports agriculture scheduling, renewable energy forecasting, and health-related humidity analysis.

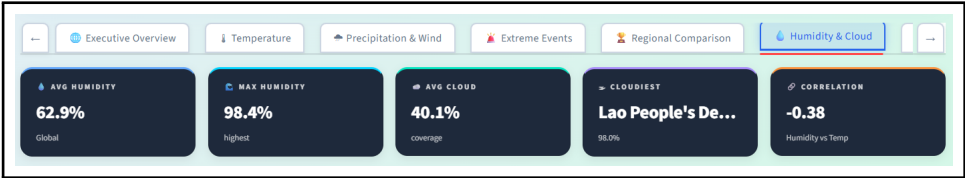


Figure 4.36: Humidity Cloud tab KPI Cards

4.6.1 Monthly Humidity Trend with Average Line

A smooth spline area chart shows the monthly humidity trend. A horizontal dashed line marks the annual mean humidity, enabling quick identification of months that are above or below average.

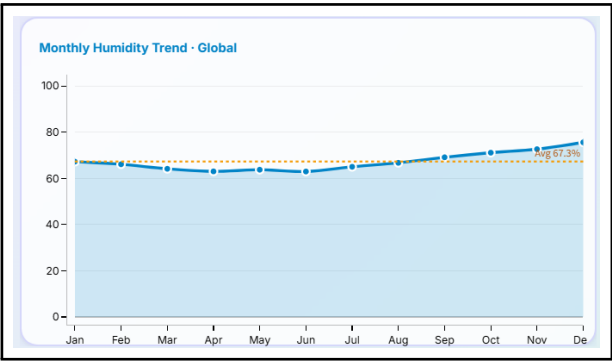


Figure 4.37: Monthly Humidity Trend with Mean Reference Line

4.6.2 Monthly Cloud Cover — Polar Bar Chart

A polar bar chart (rose chart) arranges 12 months around a circle with bar length representing average cloud cover. The colour scale transitions from light blue (low cloud) to dark navy (high cloud).

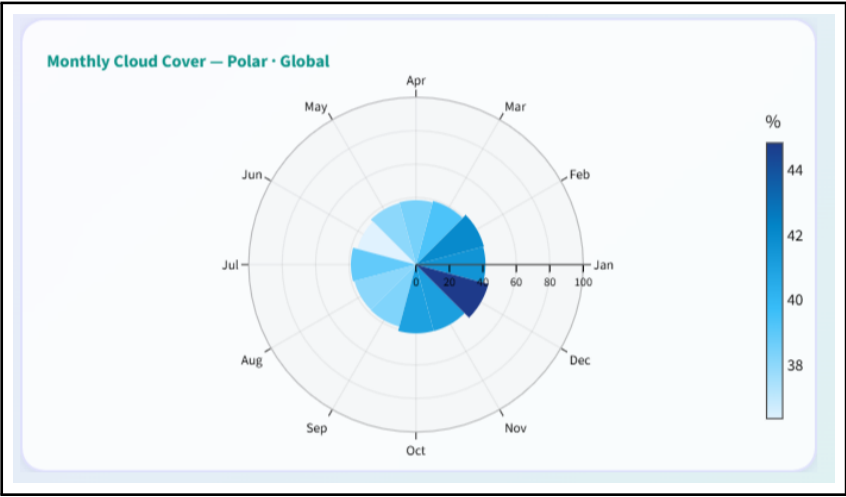


Figure 4.38: Monthly Cloud Cover Polar Bar Chart

4.6.3 Top-10 Most Humid Countries — Lollipop Chart

A lollipop chart (horizontal dot-and-line plot) ranks the ten most humid countries by mean annual humidity. A thin horizontal line extends from the zero axis to a filled circular marker at the country’s average humidity value. Value labels are positioned to the right of each dot. This chart efficiently encodes a ranked comparison with less visual clutter than a full bar chart, making the humidity differences between high-ranking tropical countries and the tenth-place country clearly visible.

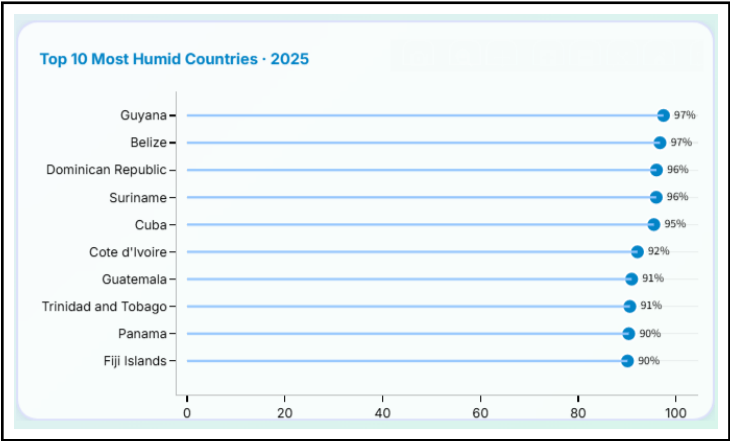


Figure 4.39: Top-10 Most Humid Countries — Lollipop Chart

4.6.4 Humidity vs Cloud Cover — Grouped Bar Chart

A grouped bar chart presents both average humidity and average cloud cover for the ten most humid countries as paired bars per country. Humidity bars are rendered in blue and cloud cover bars in purple. This chart tests the intuitive hypothesis that the most humid countries also tend to be overcast, while revealing exceptions — such as countries where high humidity coexists with low cloud cover due to ground-level moisture rather than elevated cloud formation.

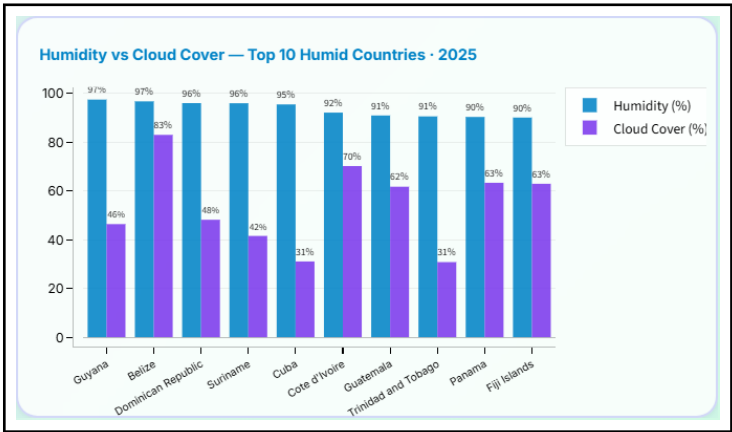


Figure 4.40: Humidity vs Cloud Cover — Grouped Bar Chart (Top 10 Humid Countries)

4.6.5 Top-10 Cloudiest Countries — Lollipop Chart

A second lollipop chart ranks the ten countries with the highest mean annual cloud cover. This complements the humidity lollipop by revealing whether the cloudiest countries overlap with the most humid ones, or whether cloud patterns are driven by different atmospheric mechanisms — such as orographic lifting along mountain ranges or persistent frontal systems at mid-latitudes.

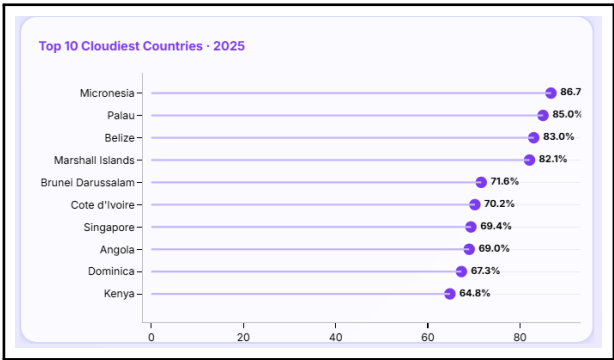


Figure 4.41: Top-10 Cloudiest Countries — Lollipop Chart

4.6.6 Humidity and Cloud Cover Dual Overlay

A dual filled-area chart overlays monthly humidity (blue) and cloud cover (purple) as smooth spline traces on a shared percentage axis. The tonexty fill between the two traces creates a visible gap band that represents months where humidity and cloud cover diverge — for example, high-humidity low-cloud months characteristic of fog or ground-level moisture conditions.

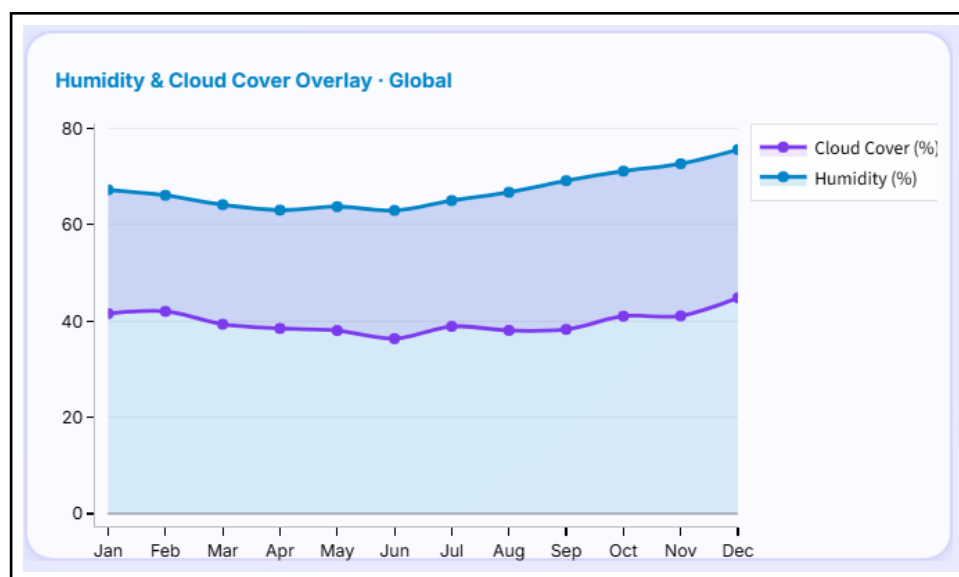


Figure 4.42: Humidity and Cloud Cover Dual Overlay Chart

4.7 UV Index & Air Quality

Tab 7 — UV Index & Air Quality: This tab addresses two variables with direct public health implications — ultraviolet radiation exposure and airborne particulate pollution. A two-panel subplot combines a monthly UV index line chart (with a dotted annual mean reference) with a radial gauge displaying the annual average UV level across Low/Moderate/High zones. A diverging bar chart compares each month's average PM2.5 concentration against the WHO guideline of $15 \mu\text{g}/\text{m}^3$, with red bars marking exceedances and green bars indicating safe months. A stacked area chart tracks both PM2.5 and PM10 across months. A dual-axis timeline overlays UV index and PM2.5 to reveal seasonal co-variation. Horizontal bar charts rank the 15 most polluted countries by both PM2.5 and PM10, colour-coded against WHO thresholds. This tab is essential for environmental health monitoring and policy benchmarking.

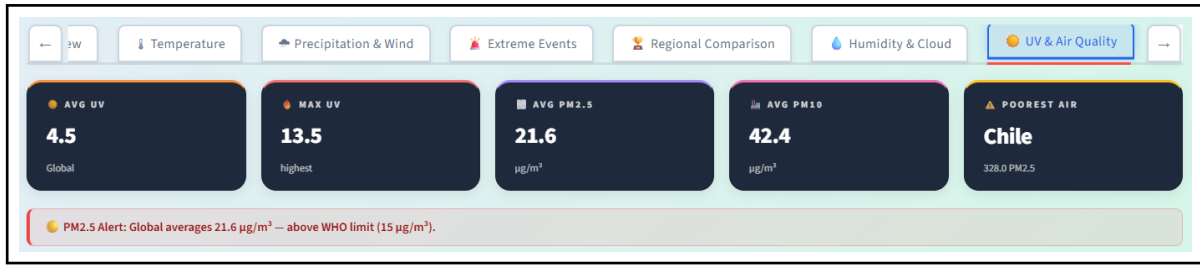


Figure 4.43: UV AQI tab KPI cards

4.7.1 UV Index Trend + Gauge Subplot

A two-panel subplot combines:

- *Left panel:* Monthly UV index line chart with average dotted reference line.
- *Right panel:* A radial gauge displaying the annual average UV, with green/yellow/red zones for Low/Moderate/High UV exposure.

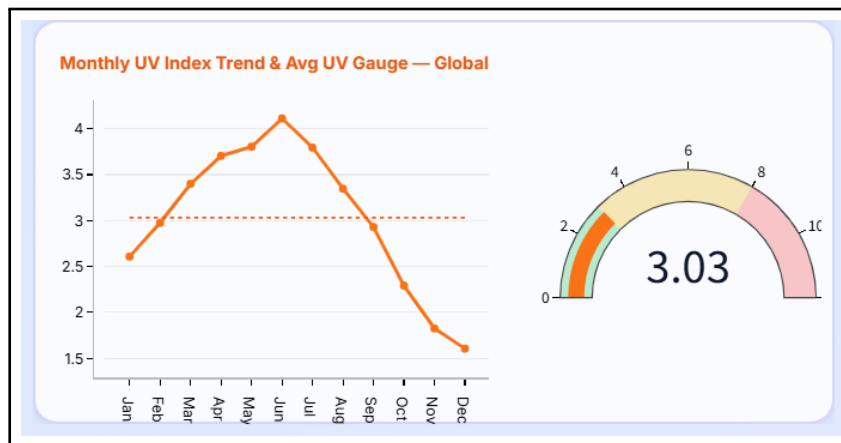


Figure 4.44: UV Index Monthly Trend and Gauge

4.7.2 PM2.5 Diverging Bar Chart vs WHO Target

Each month's average PM2.5 concentration is shown as a bar above or below the WHO guideline of $15 \mu\text{g}/\text{m}^3$. Red bars exceed the limit; green bars are within safe bounds.

PM2.5 Deviation from WHO Limit:

$$\Delta_{\text{PM}_{2.5}, m} = \overline{\text{PM}_{2.5, m}} - 15 \quad [\mu\text{g}/\text{m}^3] \quad (4.2)$$

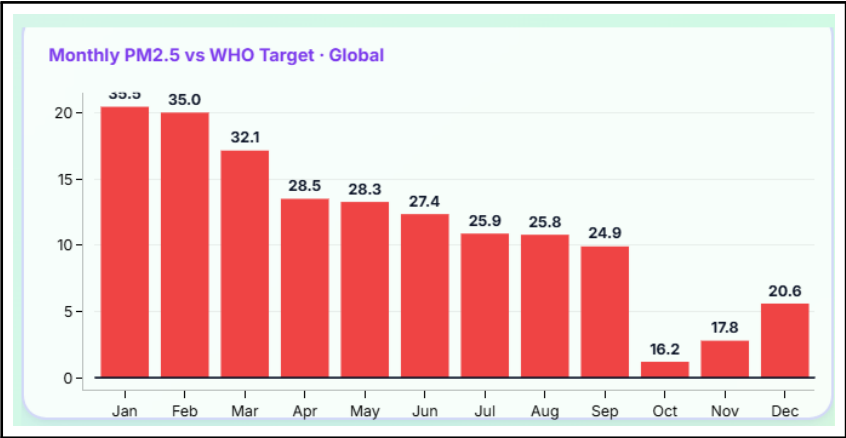


Figure 4.45: Monthly PM2.5 Diverging Bar Chart vs WHO Guideline

4.7.3 Monthly PM2.5 and PM10 — Stacked Area Chart

A layered area chart plots monthly average PM10 concentration as a bottom fill layer and PM2.5 as an overlaid fill layer on the same axis. Because PM2.5 particles are a subset of PM10, the chart visually encodes the proportional contribution of fine particles to total particulate load. A dashed horizontal reference line marks the WHO PM2.5 guideline of $15 \mu\text{g}/\text{m}^3$, enabling direct identification of months that breach safe thresholds.

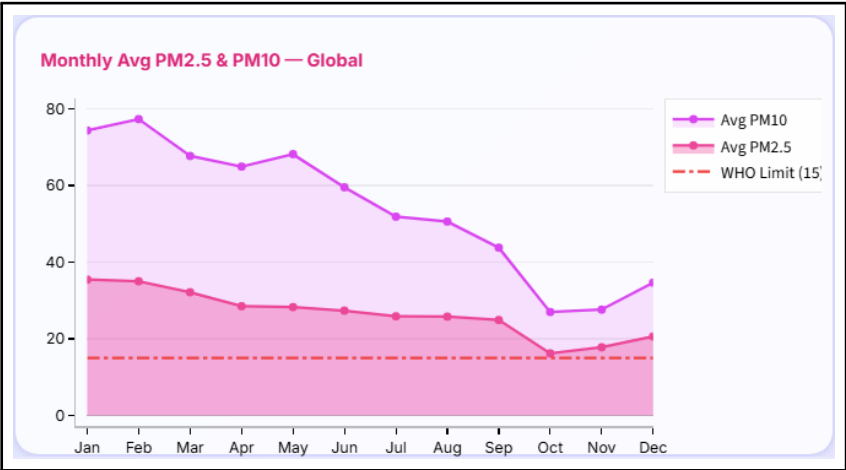


Figure 4.46: Monthly Average PM2.5 and PM10 — Stacked Area Chart

4.7.4 UV Index and PM2.5 — Dual-Axis Timeline

A dual-axis subplot places the monthly UV index (orange line) on the primary y -axis and monthly average PM2.5 (pink line) on the secondary y -axis across all twelve months. Both series use filled circular markers. This overlay exposes seasonal co-variation: in

some regions, UV index and PM2.5 peak simultaneously in dry summer months, while in others high UV occurs in a cleaner atmosphere, producing an inverse relationship.

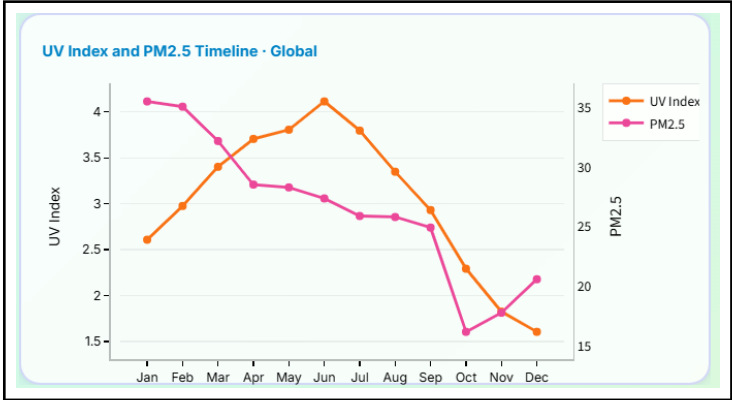


Figure 4.47: UV Index and PM2.5 — Dual-Axis Timeline

4.7.5 Top-15 Most Polluted Countries

Horizontal bar charts rank the 15 countries with the highest mean PM2.5 and PM10 concentrations. Bars are colour-coded: green (below WHO limit), amber (moderate), red (severe). A vertical dashed line marks the respective WHO guideline.

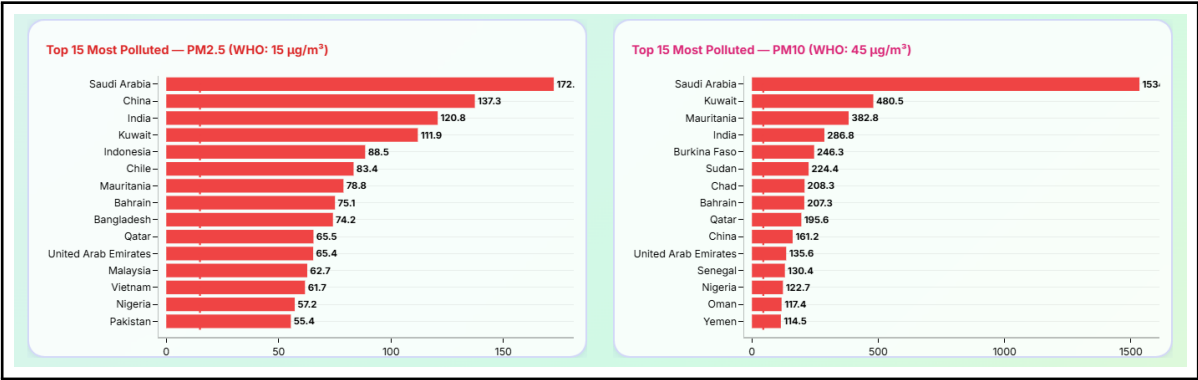


Figure 4.48: Top-15 Most Polluted Countries — PM2.5 and PM10s

4.8 Climate Risk Index

Tab 8 — Climate Risk Index: This tab synthesises all weather variables into a single composite metric — the Climate Risk Score — designed to quantify multi-hazard exposure on a 0–100 scale. An enhanced gauge indicator displays the average risk score for the selected scope and year, with the gauge title incorporating the risk category label, year-over-year trend direction, and volatility context. A

monthly risk timeline line chart tracks how the risk score evolves through the year. A treemap of the 40 highest-risk countries uses both cell area and colour intensity to convey the magnitude and hierarchy of risk at a glance. A ranked horizontal bar chart lists the top 15 risk countries explicitly. A global choropleth map closes the tab, colouring every country by its risk score on a yellow-to-dark-red scale with a gold highlight on the selected country. This tab provides the most integrated and actionable view for climate risk analysts and policymakers.

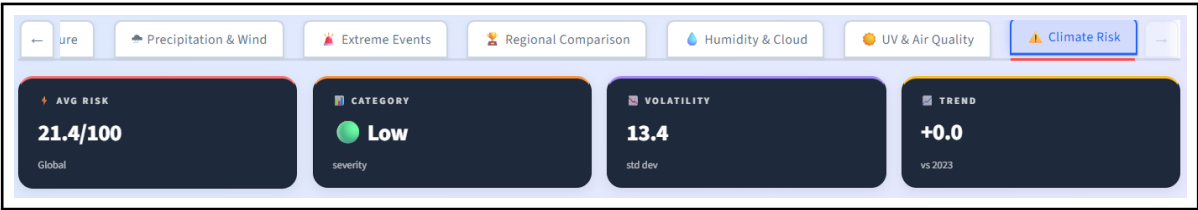


Figure 4.49: Climate risk tab KPI cards

4.8.1 Risk Score Gauge with Trend Context

An enhanced gauge indicator displays the average risk score for the selected scope and year. The title includes the risk category (Low/Medium/High), year-over-year risk trend, and volatility context.



Figure 4.50: Climate Risk Score Gauge with Full Context

4.8.2 Monthly Risk Score Timeline

A smooth spline line chart with a filled area beneath it traces the average Climate Risk Score for each month of the selected year. The y -axis is bounded at $[0, 100]$ to keep the gauge metaphor consistent with the risk score definition. Months with sustained high risk scores are immediately apparent as elevated plateaus, while rapid month-to-month transitions indicate seasonal hazard onset or cessation — for example, the beginning or end of monsoon season shifting the precipitation component of the risk formula.

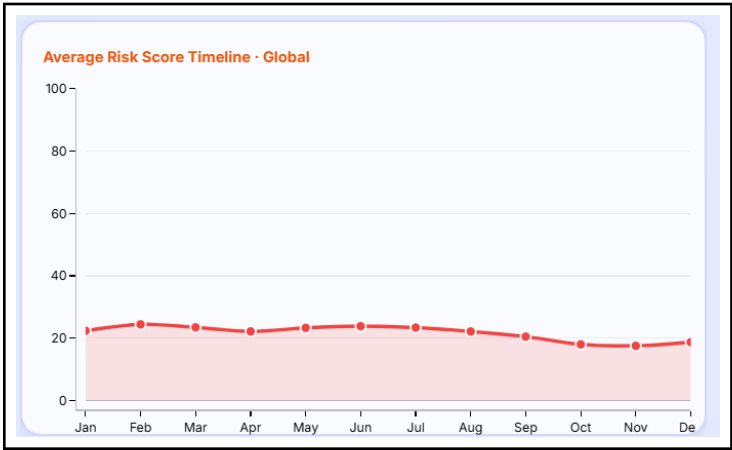


Figure 4.51: Monthly Risk Score Timeline — Spline Area Chart

4.8.3 Top-40 Risk Countries Treemap

A treemap visualises the 40 highest-risk countries; cell area and colour intensity both scale with the risk score, providing an immediate visual hierarchy of global climate danger.



Figure 4.52: Top-40 Highest Risk Countries Treemap

4.8.4 Top-15 Countries by Risk Index — Horizontal Bar Chart

A ranked horizontal bar chart lists the fifteen countries with the highest mean Climate Risk Scores. Bar fill uses the Reds continuous colour scale so that the highest-risk countries appear in the darkest red shades. Value labels are positioned outside each bar. This explicit numeric ranking complements the treemap by showing precise score differences between closely ranked countries that may appear visually similar at the treemap’s scale.

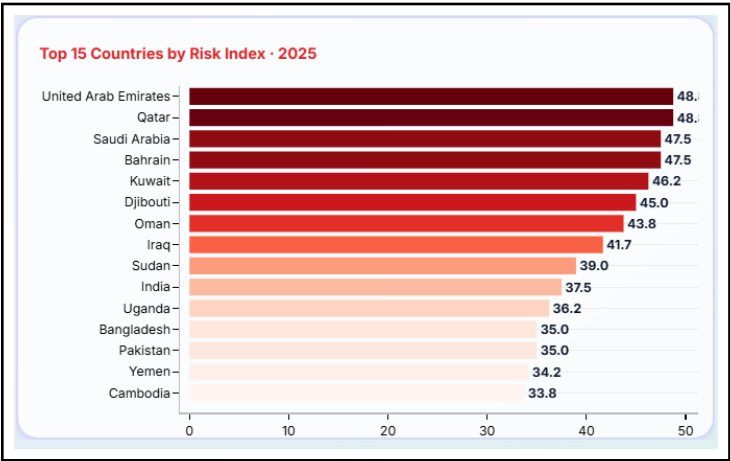


Figure 4.53: Top-15 Countries by Risk Index — Horizontal Bar Chart

4.8.5 Global Climate Risk Score Choropleth

A world choropleth map colours countries by their average Climate Risk Score. A yellow-to-dark-red colour scale conveys risk severity at a glance. The selected country is highlighted in gold.

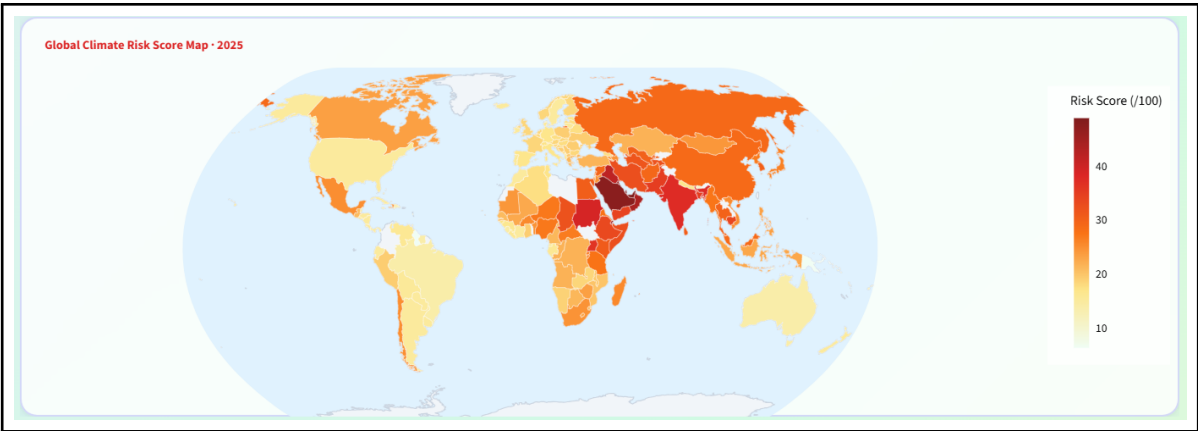


Figure 4.54: Global Climate Risk Score Choropleth Map

5. Results

5.1 Data Preparation Outcomes

Table 5.1: Data Preparation Summary

Step	Outcome
Raw dataset loaded	GlobalWeatherRepository.csv
Redundant columns removed	9 imperial/duplicate columns dropped
Missing value imputation	Median imputation (numeric), mode (categorical)
Unit conversions applied	Wind: km/h $\rightarrow$ m/s; Pressure: mb $\rightarrow$ Pa; Visibility: km $\rightarrow$ m
Datetime parsing	Year, month, day extracted
Monthly aggregation	Daily records $\rightarrow$ monthly averages per country/location/year
Output files	2 clean CSV files (daily + monthly)

5.2 Statistical Analysis Results

Table 5.2: Key Statistical Findings

Finding	Detail
Temperature distribution	Broadly distributed across climate zones; multi-modal shape reflecting tropical, temperate, and polar regions
Humidity skewness	Left-skewed; majority of records between 60%–90%
Key correlations	Temperature negatively correlated with humidity; UV index positively correlated with temperature
Seasonal pattern	Clear temperature peak in mid-year Northern Hemisphere summer months
Extreme events	Heatwave events concentrated in tropical and sub-Saharan countries; heavy rain events peak in monsoon seasons
Air quality	Several South and East Asian countries consistently exceed WHO PM2.5 guidelines
Risk leaders	Countries in tropical and arid zones consistently score highest on the composite risk index

5.3 Dashboard Features Delivered

Table 5.3: Dashboard Features Summary

Feature	Implementation
Tabs	8 themed tabs with previous/next navigation arrows
Charts	40+ interactive Plotly charts across all tabs
KPI Cards	Custom HTML/CSS cards with coloured top borders
Global Filters	Country + Year selectbox in sidebar
World Maps	4 choropleth maps (Temperature, Precipitation, PM2.5, Risk)
Risk Engine	Vectorised composite risk score formula
Caching	<code>@st.cache_data</code> on all expensive aggregations
WHO Benchmarks	PM2.5 and PM10 WHO guidelines shown as reference lines
Anomaly Alerts	Conditional alert/OK boxes based on anomaly magnitude
Country Highlight	Gold border overlay on selected country in all choropleths

6. Conclusion

6.1 Summary

This project successfully delivered a complete end-to-end data science pipeline for global weather analysis. Starting from raw, uncleaned data from the Kaggle Global Weather Repository, the work progressed systematically through three structured milestones:

- **Milestone 1** produced clean, analysis-ready daily and monthly weather datasets through careful inspection, missing value imputation, unit standardisation to SI units, and temporal aggregation.
- **Milestone 2** delivered comprehensive statistical insights into global temperature distributions, seasonal trends, country-level rankings, correlation structures, and extreme weather event detection using WMO-aligned thresholds. A novel composite Climate Risk Score was formulated to quantify multi-hazard exposure.
- **Milestone 3** culminated in a professional-grade, 8-tab interactive Streamlit dashboard with 40+ Plotly charts, real-time sidebar filters, WHO-benchmarked air quality indicators, and a global choropleth risk map.

6.2 Key Insights

The analysis revealed several important findings:

- Global temperature distributions reflect the diversity of climate zones, with strong bimodality between tropical and polar regions.
- Year-over-year temperature anomalies are detectable and quantifiable, providing an early warning signal for warming trends.
- Air quality in several densely populated South and East Asian countries significantly and consistently exceeds WHO PM2.5 guidelines.
- A small number of tropical and arid countries dominate the composite Climate Risk Index, driven by the simultaneous presence of heat, drought, and high UV exposure.

6.3 Limitations and Future Work

While the project successfully achieves its defined objectives, several limitations remain that could be addressed in future iterations.

The most significant limitation is that the dashboard operates on a **static, pre-downloaded dataset**. Because the data is loaded from a fixed CSV file, the insights reflect a historical snapshot rather than current conditions. In future work, this could be resolved by integrating a live weather API — such as OpenWeatherMap or WeatherAPI — to stream real-time observations directly into the dashboard, enabling continuous monitoring and near-real-time alerting.

A second limitation concerns **temporal granularity**. During data preparation, daily records were aggregated into monthly averages to reduce noise and improve performance. While this is appropriate for trend analysis, it smooths over short-duration extreme events such as flash floods, single-day heatwaves, or sudden wind gusts. Retaining and analysing the full daily resolution would allow the dashboard to detect event-level anomalies with far greater precision.

The **Climate Risk Score** is currently computed using a fixed set of hand-crafted weights, where each weather threshold contributes a predetermined number of points. Although these weights are grounded in meteorological reasoning, they have not been empirically validated against historical disaster or mortality data. A more rigorous future approach would be to train a supervised or unsupervised machine learning model on labelled disaster outcome data to learn optimal, data-driven risk weights that reflect real-world impact.

The current analysis operates at the **country level**, aggregating all observations within a country into a single representative value. This masks substantial within-country variation — a country as large and climatically diverse as India or the United States contains multiple distinct climate zones. Future development could incorporate sub-national breakdowns at the state, province, or city level, making the dashboard more actionable for regional planners and policymakers.

Finally, the project analyses **historical and current weather patterns** but does not extend into future climate scenarios. Incorporating projected climate data from authoritative sources such as the Intergovernmental Panel on Climate Change (IPCC) or CMIP6 model outputs would allow users to explore how temperature, precipitation, and

risk profiles are expected to evolve under different greenhouse gas emission scenarios, significantly enhancing the strategic value of the dashboard.